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LEARNER GUIDE

For

Gemini for Google Sheet

Course Code: C817

Conducted by

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Introduction

This Learner Guide accompanies Gemini for Google Sheet (C817). It follows the same three topics - Fundamentals of Google Sheet, Data Analytics and Visualization, and Interactive Dashboard - and the same twelve connected labs as the slide deck, Lesson Plan and lab folders. Every lab uses fictional data from Orchid Bay Talent Partners, a made-up Singapore recruitment and HR firm running its Q3 2026 hiring drive.

The course follows Google's official guide, Collaborate with Gemini in Google Sheets, and the AI function guide, as current in October 2026. Each concept explains what the feature is, why it matters, how it works, a worked example and when to use or avoid it. Each lab folder contains an instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook; every prompt is also on the Lab Prompts web page with a Copy button. Gemini's output remains a proposal until you have checked it in the lab's Test It step.

Course Learning Outcomes

- LO1: Use Gemini in Google Sheets - the side panel, actions and formula help - to summarise, quality-check and prepare HR data, and to generate, explain and fix formulas such as VLOOKUP, COUNTIF and nested IF.
- LO2: Create colour-coded tables, trackers and dropdowns, extract and categorise text with the AI function, and generate data analysis, insights, pivot tables and charts with Gemini, verifying every result.
- LO3: Build an interactive hiring dashboard, automate repeat screening with a custom Gem, and use Gemini's optimisation solver, reporting and sharing features responsibly.

Before You Start — Preparation

What you need

- A laptop with a current Chrome, Edge, Firefox or Safari browser.
- A Google account with Gemini in Google Sheets: a Google Workspace plan with Gemini, or a Google AI Pro or Ultra plan. The Ask Gemini (spark) button must appear at the top right of Google Sheets.
- The AI function (=AI) and Gems available on that account (Labs 6 and 11). Gemini features need a supported language - English is used throughout the course.
- Google Drive space for the twelve lab workbooks and two Google Docs.
- The course labs folder: one sub-folder per lab with its instruction PDF, sample-prompts.txt and data/ workbook (and, for Lab 11, gem-instructions.txt and two knowledge documents).

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Verify your setup

Upload labs/lab-01-*/data/OrchidBay_Q3_Applications_Export.xlsx to Google Drive, open it and choose File > Save as Google Sheets. In the new file click Ask Gemini at the top right and confirm that the side panel shows a summary of the spreadsheet. Type =AI(in an empty cell to check the AI function is recognised, and open gemini.google.com to check you can see Gems. Labels can change as Google updates the product.

```
labs/
  lab-01-<theme>/
    Lab-01-Instructions.pdf     instructions + sample prompts
    sample-prompts.txt          copy-ready prompts
    data/OrchidBay_<dataset>.xlsx
  lab-11-gems/
    gem-instructions.txt        Gem instructions to paste
    data/gem-knowledge/*.docx   job descriptions + screening rubric
Google Drive/                   your Google Sheets copies
```

Conventions used in every lab

- All Orchid Bay companies, people, IDs and values are fictional training data; emails use example.com.
- Prompts are shown in shaded boxes; copy them from the prompts web page or sample-prompts.txt and adapt them.
- Labels tell you where to send each prompt: ASK GEMINI (answer in the panel), INSERT (then Insert), ACTION (then Apply), FORMULA (then Insert into a cell), AI FUNCTION (type in a cell), GEM and GEMINI APP.
- S$ is used for currency, dates are yyyy-mm-dd and the reporting date is 2026-09-30.
- Record each prompt, what Gemini did, how you verified it and your decision on the Prompt_Log tab.
- Gemini's output can vary between runs; the Test It values come from the data, so check against them.

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Topic 01 — Fundamentals of Google Sheet (Labs 1-4)

Gemini side panel and summaries · data quality checks · data prep with Gemini actions · generating, explaining and fixing formulas (VLOOKUP, COUNTIF, nested IF)

Key concepts

- Native Sheets first — Gemini works best on a Google Sheets file - save an .xlsx as Google Sheets first.
- Ask, then insert — Answers stay in the side panel until you click Insert. Insert anything you want to keep.
- Actions you approve — For edits, Gemini shows an Action preview card. You click Apply - and can Undo.
- Name the tab and column — 'Count Hired in the Stage column on Applications' beats 'analyse my data'.
- Describe the formula — Write the rule in plain English; Gemini writes VLOOKUP, IF or COUNTIFS and explains it.
- Check every number — Confirm Gemini's answers with your own COUNTIF, COUNTIFS or pivot-table total.

Concepts for Lab 1: What Gemini in Google Sheets Can Do, Prompting Gemini and the Side Panel.

What Gemini in Google Sheets Can Do

Gemini in Google Sheets is Google's AI collaborator built into the spreadsheet. From the side panel you can create tables, create formulas, generate data analysis and insights, build charts and graphs, summarise files from Drive and emails from Gmail, and perform actions on the sheet - conditional formatting, pivot tables, dropdowns and checkboxes, sorting and filtering, find and replace, number formats, inserting, deleting or freezing rows and columns, filling ranges, formatting tables and solving optimisation tasks. Beside the panel, the AI function (=AI) brings Gemini into individual cells.

It turns spreadsheet work you would normally look up or click through into a plain-English request. That saves time on routine work, but Gemini can be wrong, so its output stays a proposal until you check it. The feature needs an eligible Google Workspace or Google AI plan and works best on native Google Sheets files.

How it works

- Open a spreadsheet in Google Sheets and click Ask Gemini (the spark icon) at the top right.
- Pick a suggested prompt or type your own in the prompt box and press Enter.
- Insert, Apply, Retry or Undo the result - then verify it against the data.

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Worked example

- Orchid Bay's analyst opens the Q3 applications export and reads Gemini's automatic summary.
- She asks which positions received the most applications and inserts the table.
- A COUNTIF confirms the Rejected count Gemini reported.

Decision guide

Use when	Avoid when
The data is approved for use with Gemini in your organisation.	You would act on an answer without checking it.
You can describe the result you want and check it afterwards.	The file is an .xlsx you have not saved as Google Sheets.

Practitioner quality lens

- Access: Eligible plan, right account, Ask Gemini visible.
- Scope: The right tabs selected as sources.
- Evidence: Every answer checked with a formula.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/13952129
- https://blog.google/products-and-platforms/products/workspace/gemini-workspace-updates-march-2026/

Prompting Gemini and the Side Panel

A prompt is the instruction you type in the side panel. Good spreadsheet prompts name the tab and columns, state the task and the output you want, and add any rules. The panel also lets you choose sources - the dropdown in the prompt box picks which tabs Gemini looks at - and offers Insert, Copy, Retry, Undo, feedback (thumbs) and More options > Clear history. Your conversation is lost when you reload, close the file or go offline.

Vague prompts give vague or wrong answers. Naming the tab and column removes guesswork, and selecting sources stops Gemini mixing up similar tables. Knowing that history disappears teaches you to insert or copy results you want to keep.

How it works

- Write: task + tab/column names + output format + rules (e.g. 'don't change anything').
- Use the sources dropdown to focus Gemini on the tabs you mean.

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- Refine with a follow-up before inserting; Retry for a different version.

Five Ways to Work with Gemini in Sheets

Aspect	Side panel answer	Insert	Action	=AI function	Gem
What you do	Ask a question	Ask, then click Insert	Ask, then click Apply	Type =AI(...) in a cell	Pick your Gem in the panel
Changes the sheet?	No	Yes, when you insert	Yes - Undo available	Yes, when you generate	Only if you insert
Best for	Questions, insights	Tables, formulas, charts	Formatting, pivots, sort/filter	Row-by-row text	Repeatable expert tasks
Lab	1, 7	1, 5, 8	2, 3, 5	6	11

Worked example

- Weak: 'analyse this'. Strong: 'Using Applications and Jobs, make a table of applications and openings per Position.'
- A follow-up adds Applications_per_Opening and sorts it before Insert.
- The result is inserted on its own tab so it survives a reload.

Decision guide

Use when	Avoid when
The task has a clear output - a number, a table, a formula or a chart.	The question needs judgement Gemini cannot see (e.g. a client's preference).
You can tell Gemini which tab and columns to use.	You rely on the panel history as your record - insert instead.

Practitioner quality lens

- Clarity: Tab, columns, task and output in every prompt.
- Focus: Sources set to the right tabs.
- Record: Results inserted; Prompt_Log updated.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/15013615

The Gemini Side Panel — What Each Part Does

- Ask Gemini (spark) — Top right of Sheets. Opens the panel with a summary of the sheet.
- Suggestions and Gems — Suggested prompts (More suggestions for others) and premade or custom Gems.

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- Prompt box and sources — Type a prompt; the sources dropdown picks which tabs Gemini uses.
- Response controls — Insert, Copy, Retry, Preview, Analysis steps, Export to Docs, thumbs.
- More options, Expand, Close — Clear history; make the panel larger; close it.

Shortcuts and Key Controls

- Insert / Copy — Keep a result: insert it into the sheet or copy it.
- Retry — A different version; Retry with Google Search adds the web.
- Analysis steps / Show code — See how Gemini calculated an answer.
- Settings / Undo — Adjust or reverse the last action.
- Shortcuts — Summarise: Cmd+Ctrl+N / Ctrl+Alt+N. Formula: = then Cmd+Ctrl+G / Ctrl+Alt+G.

Lab 1 · Gemini Side Panel & Summaries — Open Gemini and Explore the Q3 Candidate Export

Learning outcome: LO1: open and use the Gemini side panel in Google Sheets to summarise a spreadsheet, ask questions about HR data across tabs and insert results, checking each answer against the data.

Goal: Get to know Gemini in Google Sheets on a real recruitment export: summarise it, question it across two tabs, insert a result and spot what looks wrong.

You are a talent analyst at Orchid Bay Talent Partners, a fictional Singapore recruitment and HR firm. The Q3 2026 hiring drive has just closed its first round: 243 rows exported from the applicant tracking system for eight client roles. You will open the Gemini side panel, read its automatic summary, ask questions in plain English, focus Gemini on the tabs you choose, insert a table into the sheet, and try the panel's controls - Retry, Undo, feedback and Clear history.

What you'll build

OrchidBay_Q3_Applications_Export as a Google Sheets file with an inserted Applications_by_Position table, a check formula that confirms Gemini's stage count, and a filled Prompt_Log (Tools: Google Sheets · Gemini side panel (summary, prompts, sources, Insert) · OrchidBay_Q3_Applications_Export.xlsx.)

Prerequisites

- A Google account with Gemini in Google Sheets: a Google Workspace plan with Gemini, or a Google AI Pro / Ultra plan. The Ask Gemini (spark) button must be visible at the top right of Sheets.
- Google Chrome or another current browser, signed in to that account.

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- This lab folder downloaded, including data/OrchidBay_Q3_Applications_Export.xlsx.
- Data handling: all Orchid Bay data is fictional. With real candidate data, use only tools your organisation has approved, and only the columns the task needs.

Step-by-step

1. Download OrchidBay_Q3_Applications_Export.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Q3_Applications_Export (Google Sheets copy,
not .xlsx)

Tabs      : About_This_Data | Applications | Jobs | Prompt_Log

Gemini    : side panel open on the right
```

2. When the side panel opens on a sheet that has data, Gemini shows a short summary of the spreadsheet and some suggested prompts. Click View more to read all of it. You can summarise at any time with the shortcut Cmd + Ctrl + N (Mac) or Ctrl + Alt + N (Windows). Now ask in your own words. Gemini should say that one row is one candidate application, received between 2026-07-01 and 2026-09-26.

```
ASK GEMINI:

Summarize this spreadsheet. What does one row on the Applications tab
represent, and what time period does it cover?
```

3. Ask Gemini to describe the columns, and compare its answer with the data dictionary on About_This_Data. It should report 243 rows of data and 16 columns. If a number is different, ask Gemini how it counted (for example, whether it included the header row).

```
ASK GEMINI:

Describe each column on the Applications tab in one line, and tell me how
many rows of data there are.
```

4. Gemini can read every tab. To focus it, open the sources dropdown in the prompt box and tick only Applications and Jobs, then ask a question that needs both tabs. Before you insert the table, refine it with a follow-up. Data Analyst should show 41 applications for 2 openings. Click Insert, then rename the new tab Applications_by_Position. Insert keeps the result: the side-panel history is lost when you reload the page or go offline.

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```
INSERT:

Using the Applications and Jobs tabs, make a table of each Position with
the number of applications and the number of Openings.


FOLLOW-UP:

Add a column Applications_per_Opening and sort the table from highest to
lowest.
```

5. Ask about the pipeline. Gemini should name Rejected as the largest stage (150 rows) and report 11 Hired. Then type the check formula into an empty cell on the Applications_by_Position tab - it must match Gemini's Rejected count. If it does not, the formula wins: ask Gemini how it counted.

```
ASK GEMINI:

How many candidates are at each Stage? Which stage has the most candidates?


CHECK:

Rejected count   =COUNTIF(Applications!O2:O244,"Rejected")

Hired count      =COUNTIF(Applications!O2:O244,"Hired")
```

6. Explore the panel's controls. Send the first prompt, then click Retry for a different version. Use the thumbs up or down to give feedback. Open More options (three dots) to see More suggestions and Clear history, and Expand to make the panel larger. Then use Gemini for help with Sheets itself - you will use the answer about duplicates in Lab 3.

```
ASK GEMINI:

Explain this spreadsheet like I'm 5 years old.


ASK GEMINI:

How can I remove duplicates in Google Sheets?
```

7. Finish by asking Gemini what looks wrong - do not fix anything yet. A good answer mentions several of: repeated Candidate_IDs, Source spelt more than one way (e.g. 'Linked In'), blank emails, salaries typed as text such as 'S$4,800', and a negative Years_Experience. Lab 2 checks each one.

```
ASK GEMINI:

Is there anything in the Applications tab that looks wrong or inconsistent?
List the issues with an example of each. Don't change anything.
```


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8. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```

Sample prompts

Purpose	Prompt
Summary	Give me a five-bullet summary of this candidate data for the Head of Talent.
Question	Which Source brought in the most applications?
Question	Which recruiter owns the most applications, and for which positions?
Help	How do I freeze the header row in Google Sheets?
Insight	What's interesting about this data? Give three observations with the numbers behind them.

Test it

Check	Expected result
Rows and columns	243 rows of data, 16 columns; dates 2026-07-01 to 2026-09-26.
Applications_by_Position tab	8 positions; Data Analyst 41 applications for 2 openings; sorted by Applications_per_Opening.
Stage check	Rejected = 150 and Hired = 11, matching Gemini's answer.
Issues spotted	Gemini lists repeated IDs, Source spelling variants, blank or invalid emails, text salaries and a negative Years_Experience.
Prompt_Log	Every prompt you ran has What Gemini did, How I verified and a Decision.

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Checkpoint and rejoin point

Keep this file in Drive. Lab 2 opens its own copy of the same export, so you can start Lab 2 even if this lab is not finished.

Troubleshooting

If this happens	Fix
There is no Ask Gemini (spark) button.	Gemini needs an eligible Google Workspace or Google AI plan. Check you are signed in to the right account (profile picture, top right) and that your admin has turned Gemini on. Pair with a neighbour meanwhile.
Gemini says it can't help with this file.	You are probably in the .xlsx file. Choose File > Save as Google Sheets and work in the new copy.
Gemini's row count is off by one.	It may have counted the header row. Ask: 'Count only data rows, excluding the header.'
The side-panel conversation disappeared.	History is lost when you reload, close the file or go offline. Insert results you want to keep.
Gemini answers about the wrong tab.	Use the sources dropdown in the prompt box to select only the tabs you mean, and name the tab in the prompt.

Challenge

Ask Gemini which recruiter has the highest share of Rejected applications. Confirm it with two COUNTIFS formulas for that recruiter.

Reflection

Which of Gemini's answers would you have trusted without the COUNTIF check, and why is that risky?

Note: The lab folder labs/lab-01-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Q3_Applications_Export.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Concepts for Lab 2: Why Data Quality Matters.

Why Data Quality Matters

Data quality is whether data is complete, consistent, unique, valid and accurate enough for the decision it supports. In recruitment exports the usual problems are duplicate application IDs, the same source spelt several ways ('LinkedIn', 'Linked In', 'JobStreet ' with a trailing space), missing or invalid emails, numbers stored as text ('S$4,800') and impossible values such as negative experience.

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Every count, chart and AI answer inherits the errors in its source. Duplicates inflate applications, spelling variants split a source into several, and a salary typed as text is silently ignored by AVERAGE. Gemini analyses the data you give it - it does not make bad data good.

How it works

- Profile first: rows vs unique IDs, distinct values per category column, blanks and data types.
- Flag problems with conditional formatting before changing anything.
- Prove each count with a formula (COUNTUNIQUE, COUNTBLANK, COUNTIF).

Worked example

- 243 rows but 240 unique Candidate_IDs: 3 duplicates.
- 'Linked In', 'linkedin' and 'JobStreet ' would split two sources.
- 5 salaries typed as text and one negative Years_Experience.

Decision guide

Use when	Avoid when
Before any analysis, dashboard or AI insight on new data.	Fixing a value would mean guessing - flag it and ask the owner.
Data comes from an export or several systems.	Deleting rows you cannot explain.

Practitioner quality lens

- Unique: Rows = unique IDs, or duplicates explained.
- Consistent: One spelling per category.
- Valid: Numbers are numbers, dates are dates, values in range.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/78413

Lab 2 · Data Quality Check — Check the Quality of the Candidate Data with Gemini

Learning outcome: LO1: use Gemini to profile HR data for duplicates, inconsistent categories, missing or invalid values and wrong data types, highlight the problems and confirm each count with a formula.

Goal: Find, highlight and count every data-quality problem in the Q3 export before anyone analyses it - and prove each count with a formula.

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Before the recruitment manager sees any numbers, you check the raw Q3 export. You will ask Gemini to profile the data, find the repeated Candidate_IDs and the Source spelling variants, and apply conditional formatting actions that highlight duplicates, bad emails, text salaries and impossible values. A Checks tab of formulas confirms every count, and Gemini drafts a short data-quality report. You do not fix anything yet - that is Lab 3.

What you'll build

OrchidBay_Q3_Applications_QualityCheck with duplicates and problem cells highlighted, a Checks tab of formulas confirming each issue count, an inserted data-quality report and a filled Prompt_Log (Tools: Google Sheets · Gemini side panel (questions, conditional-formatting actions, formulas) · OrchidBay_Q3_Applications_QualityCheck.xlsx.)

Prerequisites

- Lab 1 completed, or at least the Gemini side panel opened once.
- This lab folder downloaded, including data/OrchidBay_Q3_Applications_QualityCheck.xlsx.

Step-by-step

1. Download OrchidBay_Q3_Applications_QualityCheck.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Q3_Applications_QualityCheck (Google Sheets copy)

Tabs      : About_This_Data | Applications | Jobs | Checks | Prompt_Log

Rule      : find and flag only - do not fix anything in this lab
```

2. Ask Gemini to profile the data for the common problems. Read its list carefully: it is a starting point, not the final answer - the formulas in Step 6 decide.

```
ASK GEMINI:

Check the Applications tab for data-quality problems: duplicate
Candidate_IDs, the same Source spelt different ways, missing or invalid
emails, Expected_Salary values stored as text, and impossible values.

Don't change anything. List each problem with how many rows it affects.
```

3. Ask about duplicates, then apply Gemini's conditional-formatting action. An Action preview card appears - read it and click Apply. When it finishes, Settings and Undo

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appear: Settings opens the rule, Undo reverses it. There are 243 rows but only 240 unique IDs: CAN-26531, CAN-26601, CAN-26608 each appear twice, so 6 cells turn orange (press Ctrl + Down arrow to reach the copies near the end).

```
ASK GEMINI:

Which Candidate_IDs appear more than once? Are those rows exact copies of
each other?


ACTION:

Highlight duplicate values in the Candidate_ID column in orange.
```

4. Ask Gemini to list Source exactly as spelt. Besides the seven real sources you should see 'Linked In' (3), 'linkedin' (2) and 'JobStreet ' with a trailing space (3). Spelling variants split one source into several and corrupt every count by Source.

```
ASK GEMINI:

List every distinct value in the Source column exactly as it is spelt,
including spaces and capital letters, with a count of each.
```

5. Send these one change at a time. After each Action preview card, click Apply, look at the sheet, and use Undo if it is not what you wanted. Expect 6 email cells (4 blank, 2 without @), 5 text salaries and 1 negative Years_Experience (CAN-26607).

```
ACTION:

1) Highlight cells in the Email column that are blank or do not contain @
in light red.

2) Highlight Expected_Salary cells that contain text instead of a number in
yellow.

3) Highlight Years_Experience values below 0 in red.
```

6. Now prove each count. The Checks tab already has the labels in column A and the expected values in column C. First ask Gemini for the unique-ID formula and insert it into Checks!B3. Then paste the other check formulas into column B beside their labels. Every B must match its C; if Gemini's Step 2 numbers disagree, the formulas win.

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```
FORMULA:

Create a formula that counts the unique Candidate_IDs in Applications
column A, rows 2 to 244.



CHECK:

Total rows            =COUNTA(Applications!A2:A244)

Unique IDs            =COUNTUNIQUE(Applications!A2:A244)

Blank emails          =COUNTBLANK(Applications!C2:C244)

Emails without @      =COUNTIFS(Applications!C2:C244,"<>",Applications!
C2:C244,"<>*@*")

Text salaries         =SUMPRODUCT(--ISTEXT(Applications!K2:K244))

Negative experience   =COUNTIF(Applications!H2:H244,"<0")
```

7. Ask Gemini for a short report for the recruitment manager, then click Insert - Gemini adds the table to your spreadsheet (usually on a new tab); rename that tab DQ_Report. Or use Copy and paste it into an email draft.

```
INSERT:

Write a short data-quality report for the recruitment manager: each issue,
how many rows it affects, why it matters for screening, and the fix you
recommend. Use a table.
```

8. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```


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Sample prompts

Purpose	Prompt
Profile	Which columns have blank cells, and how many blanks does each have?
Validate	Are there any Applied_Date values outside July to September 2026?
Validate	Do any rows have a Position that doesn't match the Position for their Job_ID on the Jobs tab?
Highlight	Highlight Full_Name cells that are all in capital letters in light purple.
Explain	Why is a salary stored as text a problem when I calculate an average?

Test it

Check	Expected result
Duplicates	243 rows, 240 unique IDs; CAN-26531, CAN-26601, CAN-26608 each appear twice; 6 orange cells.
Source variants	'Linked In' 3, 'linkedin' 2, 'JobStreet ' (trailing space) 3.
Emails	4 blank + 2 without @ (CAN-26697, CAN-26606) = 6 light-red cells.
Salaries and experience	5 text salaries in yellow; 1 negative Years_Experience (CAN-26607) in red.
Checks tab	Six formulas in column B match the expected values already in column C (243, 240, 4, 2, 5, 1).

Checkpoint and rejoin point

Lab 3 opens a fresh copy of the same raw export and fixes every issue you found here.

Troubleshooting

If this happens	Fix
There is no Apply button - Gemini only explains how.	Rephrase as a direct instruction ('Highlight ... in orange'), and make sure the Applications tab is selected. If actions are not available on your plan, use Format > Conditional formatting with Gemini's steps.
Gemini highlights the whole row instead of the cell.	Click Undo and say 'only the cells in the Candidate_ID column'.
COUNTUNIQUE returns 243.	Check the range: it must be A2:A244 on Applications, not a range that includes the header twice.
Gemini's counts differ from the formulas.	Ask Gemini how it counted. Trust the formula and refine the prompt.

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Challenge

Add a formula that counts the rows whose Position does not match the Position of their Job_ID on the Jobs tab (hint: ask Gemini for a SUMPRODUCT with VLOOKUP). The answer should be 0.

Reflection

Which issue would most distort a 'hires by source' report if nobody had caught it?

Note: The lab folder labs/lab-02-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Q3_Applications_QualityCheck.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Concepts for Lab 3: Gemini Actions: Preview, Apply, Undo, Data Prep: From Export to Clean Table.

Gemini Actions: Preview, Apply, Undo

An action is a change Gemini makes to your sheet from a prompt: apply conditional formatting, create a pivot table, add a dropdown or checkbox, sort, filter or clear a filter, find and replace, set a number format, insert, delete or freeze rows and columns, fill a range or format a table. Gemini first shows an Action preview card; nothing changes until you click Apply. Afterwards Settings (when available) lets you adjust the action and Undo reverses it. One prompt can combine several actions - a multi-step task.

Actions turn menus you would hunt for into one sentence, while the preview keeps you in control. Because Undo is available only until you make further changes, you check each action straight away.

How it works

- Describe the change, naming the tab and column ('In the Source column only ...').
- Read the Action preview card, then click Apply.
- Check the sheet immediately; use Settings to adjust or Undo to reverse.

Worked example

- 'Replace Linked In and linkedin with LinkedIn in the Source column' - Apply - 7 sources remain.
- 'Delete the Internal_Notes column' removes notes screening does not need.
- A multi-step prompt formats dates, adds a Stage dropdown and highlights Hired rows.

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Decision guide

Use when	Avoid when
The change is clear and you can see the result on the sheet.	The change cannot be checked or undone easily (e.g. deleting rows on the only copy).
You want several routine edits in one go.	The instruction is ambiguous about which column or range.

Practitioner quality lens

- Preview: Action card read before Apply.
- Control: Checked straight away; Undo or Settings used.
- Count: Row count recorded before and after.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/14239833

Data Prep: From Export to Clean Table

Data preparation turns a raw export into a table you can analyse: one header row, one record per row, one kind of value per column, consistent categories, real numbers and dates, and no notes or blank rows inside the data. In Google Sheets you combine Gemini actions with built-in tools such as Data > Data cleanup > Remove duplicates and Trim whitespace, then format the range as a table.

Clean data makes every later formula, pivot table, chart and AI answer trustworthy. Fixing in helper columns and keeping counts before and after lets you prove nothing was lost.

How it works

- Remove exact duplicates and trim whitespace with Data cleanup.
- Standardise categories and convert text numbers with Gemini actions.
- Tidy structure: helper columns, drop unneeded columns, freeze headers, apply a table style.

Worked example

- 243 rows become 240 after Remove duplicates.
- Name_Clean uses =PROPER(TRIM(B2)); the original names stay.
- A Stage dropdown stops new spelling variants creeping in.

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Decision guide

Use when	Avoid when
Before analysis, dashboards or sharing.	The original export is your only copy - work on a copy.
The same export arrives every week - write the steps down once.	A fix needs information you don't have.

Practitioner quality lens

- Shape: One header row, one record per row.
- Types: COUNT over a number column equals the row count.
- Audit: Originals kept; counts before and after.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/14239833
- https://support.google.com/docs/answer/186103

Lab 3 · Data Prep — Clean and Prepare the Candidate Data with Gemini Actions

Learning outcome: LO1: prepare raw HR data with Gemini actions and Sheets data-cleanup tools - remove duplicates, find and replace, set number formats, insert and delete columns, freeze, sort, filter and format as a table.

Goal: Turn the raw Q3 export into a clean, analysis-ready table of 240 applications, keeping a count before and after every change.

You fix everything you found in Lab 2. Gemini explains how to remove duplicates, then performs actions for you: find and replace the Source variants, convert text salaries to numbers with a currency format, add a clean name column, delete the notes column, freeze the headers, apply a table style, and sort and filter the data. A multi-step prompt finishes the clean-up in one go. You check the row count after every change.

What you'll build

OrchidBay_Q3_Applications_Prep as a clean table: 240 rows, seven Source values, numeric S$ salaries, a Name_Clean column, frozen headers, a Stage dropdown and Hired rows highlighted (Tools: Google Sheets · Gemini actions (find and replace, number format, insert/delete column, freeze, sort, filter, table style) · Data cleanup.)

Prerequisites

- Lab 2's list of issues (or its Test It values) to hand.
- This lab folder downloaded, including data/OrchidBay_Q3_Applications_Prep.xlsx.

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Step-by-step

1. Download OrchidBay_Q3_Applications_Prep.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Q3_Applications_Prep (Google Sheets copy)

Tabs      : About_This_Data | Applications | Jobs | Prompt_Log

Start     : 243 rows, 17 columns (including Internal_Notes)
```

2. Ask Gemini how to remove duplicates, then follow its steps: select the whole Applications data range (click A1, then Ctrl + A), choose Data > Data cleanup > Remove duplicates, tick 'Data has header row' and all columns. Sheets should report 3 duplicate rows removed, 240 rows remain.

```
ASK GEMINI:

How can I remove the duplicate rows on the Applications tab and keep one
copy of each?
```

3. Fix the Source spellings with a Gemini action and click Apply on the preview card. Then remove the trailing space from 'JobStreet ' and the double spaces in names: select columns B and G and choose Data > Data cleanup > Trim whitespace. Open the filter on Source (Data > Create a filter) - you should see exactly 7 values.

```
ACTION:

In the Source column, find 'Linked In' and 'linkedin' and replace them with
'LinkedIn'.
```

4. Fix the data types and the impossible value one change at a time, clicking Apply after each preview. Afterwards =COUNT(K2:K241) must return 240 - every salary is a number. The recruiter confirmed that CAN-26607 has 3 years of experience.

```
ACTION:

1) In the Expected_Salary column, find 'S$' and ',' and replace them with
nothing.

2) Set the number format of the Expected_Salary column to currency S$ with
no decimal places.

3) For Candidate_ID CAN-26607, change Years_Experience from -3 to 3.
```

5. Tidy the structure. The new Name_Clean column fixes names typed in capitals or with extra spaces; check that it is filled all the way to the last row. Then remove the

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notes column - screening does not need free-text notes about candidates - and freeze the headers.

```
ACTION:

1) Insert a new column named Name_Clean to the right of Full_Name and fill
it with =PROPER(TRIM(B2)) down to the last row.

2) Delete the Internal_Notes column.

3) Freeze the top row and the first column.
```

6. Apply a table style, then sort and filter. Gemini's sort and filter apply to the active table. Note the filtered count - it should be 2 rows - then clear the filter so all 240 rows show again. If the table style is not applied, select the data and choose Format > Convert to table.

```
ACTION:

1) Apply a table style to the data on the Applications tab to make it
easier to read.

2) Sort the table by Applied_Date in descending order.

3) Filter for Software Engineer candidates whose Source is Referral.

4) Clear the filter so all rows are visible.
```

7. Finish with one multi-step prompt - Gemini combines several actions in a single request. Check each part: dates show as yyyy-mm-dd, Stage has a dropdown, and 11 Hired rows are light green.

```
ACTION:

Finish cleaning this table: format Applied_Date as yyyy-mm-dd, make the
Stage column a dropdown with Applied, Screened, Shortlisted, Interviewed,
Offered, Hired, Rejected and Withdrawn, and highlight in light green the
rows where Stage is Hired.
```

8. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```


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Sample prompts

Purpose	Prompt
Fill	Fill the empty Email cells with the text 'missing - ask candidate'.
Insert	Insert a new column called Source_Group after Source and leave it empty.
Number format	Format Screening_Score as a whole number and Interview_Score with one decimal place.
Delete rows	Delete the rows where Stage is Withdrawn.
Help	How do I split the Key_Skills column into separate columns?

Test it

Check	Expected result
Rows	240 rows after removing 3 duplicates.
Source	Exactly 7 values in the filter; LinkedIn 49, JobStreet 50.
Salaries	=COUNT over the Expected_Salary column returns 240; values show as S$ with no decimals.
Structure	Name_Clean filled to the last row; Internal_Notes gone; headers frozen; CAN-26607 shows 3 years.
Sort, filter, dropdown	Newest date (2026-09-26) at the top; the filter showed 2 rows; Stage dropdown added; 11 Hired rows light green.

Checkpoint and rejoin point

This clean table is the starting point for Lab 4, which supplies the same 240 rows (plus a Budget_Max column) in its own workbook.

Troubleshooting

If this happens	Fix
Salaries still align left after the find and replace.	They are still text. Select the column and choose Format > Number > Number, or ask Gemini: 'Convert the Expected_Salary values to numbers.'
Gemini replaced 'LinkedIn' everywhere in the sheet.	Click Undo and name the column: 'In the Source column only ...'.
Name_Clean stops part-way down.	Ask Gemini to 'fill the formula down to row 241', or double-click the fill handle on C2.
No action card appears for sort, filter or table style.	Do it from the menu: Data > Sort range, Data > Create a filter, Format > Convert to table.
Sort mixed up the rows.	Undo, then make sure the whole table is selected (or converted to a table) before sorting.
Remove duplicates removed 0 rows.	You selected only part of the data. Select from A1 to the last row and column, then run it again.

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Challenge

Ask Gemini to add a Source_Group column that labels LinkedIn, JobStreet, MyCareersFuture and Indeed as 'Job board' and the rest as 'Direct'. Check the two groups add up to 240.

Reflection

Why should you fix data in a helper column, or keep the original, rather than overwrite it?

Note: The lab folder labs/lab-03-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Q3_Applications_Prep.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Concepts for Lab 4: Formulas: VLOOKUP, COUNTIF and Nested IF, Generate, Explain and Fix Formulas with Gemini.

Formulas: VLOOKUP, COUNTIF and Nested IF

A formula starts with = and calculates a value. VLOOKUP(search key, range, column number, FALSE) finds a key in the first column of a range and returns a value from another column. IF(test, if true, if false) returns one of two results, and IFs can be nested, with AND/OR, for three or more outcomes. COUNTIF counts cells that meet one condition; COUNTIFS counts rows that meet several.

Formulas make screening rules repeatable and auditable: anyone can click a cell and see the rule, and the result recalculates when data or thresholds change. Keeping thresholds in a Settings tab means one edit re-screens everyone.

How it works

- Write the business rule in words, including edge cases.
- Lock lookup ranges and setting cells with $ before filling down.
- Check one row by hand and reconcile totals with COUNTIF.

Worked example

- =VLOOKUP(E2,Jobs!$A$2:$H$9,5,FALSE) returns the role's Min_Years.
- =IF(K2<=Q2,"Within budget","Above budget") checks salary fit.
- Strong + Possible + Not suitable = 240.

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Decision guide

Use when	Avoid when
A value follows a fixed rule from other values.	Typing a number that could be calculated.
The result must update when data or thresholds change.	Hard-coding a threshold inside the formula.

Practitioner quality lens

- Readable: Rule in words matches the formula.
- Robust: Ranges locked; errors handled.
- Reconciled: One row by hand; totals add up.

Authoritative references

- https://support.google.com/docs/answer/3093318
- https://support.google.com/docs/answer/3093364
- https://support.google.com/docs/answer/3256550

Generate, Explain and Fix Formulas with Gemini

Gemini creates formulas from a description: ask in the side panel, or type = in a cell and press Cmd + Ctrl + G (Mac) or Ctrl + Alt + G (Windows), then click Insert into the selected cell. It can explain any formula step by step. When a cell shows an error, hover over it and click Fix: Gemini reviews the formula, explains the problem and suggests a correction.

You no longer need to remember syntax - but you still need to read the formula. Explanations teach as you go and help you audit inherited sheets; Fix turns cryptic errors such as #REF! into plain English.

How it works

- Click the target cell, describe the formula with cell references, then Insert and fill down.
- Ask Gemini to explain the formula and work through one row.
- Hover over an error and click Fix; read why before accepting.

Worked example

- #REF! in Budget_Max: VLOOKUP asked for column 9 of an 8-column range - Fix changes it to 8.
- Gemini writes the nested IF for Screen_Result from one sentence.
- Its explanation of row 2 matches M2, H2 and R2.

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Decision guide

Use when	Avoid when
You know the rule but not the syntax.	You can't read the formula Gemini wrote - ask for an explanation first.
You inherit a sheet full of formulas you didn't write.	The fix changes the meaning of the rule, not just the syntax.

Practitioner quality lens

- Placement: Target cell selected before Insert.
- Understanding: Explanation matched to real values.
- Fixes: Reason for the error understood.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/3093318

Lab 4 • Formulas: VLOOKUP, COUNTIF & Nested IF — Generate, Explain and Fix Screening Formulas

Learning outcome: LO1: use Gemini to generate VLOOKUP, IF, nested IF and COUNTIF/COUNTIFS formulas from plain-English rules, explain them step by step and fix formula errors, verifying the results on one row and in total.

Goal: Add four screening columns to the clean applications - Budget_Max, Min_Years, Salary_Fit and Screen_Result - and prove they are right.

The hiring managers want a first screen of all 240 applications. You will fix a broken lookup with Gemini's Fix button, then describe each rule in plain English and let Gemini write the formula: a VLOOKUP for the role's minimum experience, an IF for salary fit, and a nested IF with AND that labels each candidate Strong, Possible or Not suitable. Gemini explains the formula, COUNTIF and COUNTIFS summarise the results, and independent check formulas prove them.

What you'll build

OrchidBay_Screening_Formulas with working Budget_Max, Min_Years, Salary_Fit and Screen_Result columns, a Checks tab of COUNTIF/COUNTIFS results that reconcile to 240, and a filled Prompt_Log (Tools: Google Sheets · Gemini formula help (side panel, = then Cmd/Ctrl + Ctrl/Alt + G, Fix) · OrchidBay_Screening_Formulas.xlsx.)

Prerequisites

- Labs 1-3 completed, or familiarity with the Applications columns (see About_This_Data).

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- This lab folder downloaded, including data/OrchidBay_Screening_Formulas.xlsx.

Step-by-step

1. Download OrchidBay_Screening_Formulas.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Screening_Formulas (Google Sheets copy)

Tabs      : About_This_Data | Applications | Jobs | Settings | Checks |
Prompt_Log

Settings  : B1 report date 2026-09-30 | B2 pass mark 70 | B3 possible mark
60
```

2. Column Q (Budget_Max) shows an error. Hover over Q2 and click Fix: Gemini reviews the formula and explains the problem - VLOOKUP asks for column 9 of a range that has only 8 columns. Accept the fix, then copy Q2 down to Q241 (or double-click the fill handle). Q2 should show S$6,500. If you don't see Fix, ask Gemini instead.

```
Before    : =VLOOKUP(E2,Jobs!$A$2:$H$9,9,FALSE)  ->  #REF!

After     : =VLOOKUP(E2,Jobs!$A$2:$H$9,8,FALSE)  ->  S$6,500


ASK GEMINI:

Explain why the formula in Q2 shows #REF! and give me the corrected
formula.
```

3. Type Min_Years in R1. Click R2, then ask Gemini for the formula - in the side panel, or type = in the cell and press Cmd + Ctrl + G (Mac) / Ctrl + Alt + G (Windows). Click Insert, then fill it down to row 241. R2 should show 2 (Data Analyst).

```
FORMULA:

Create a formula for R2 that looks up the Job_ID in E2 on the Jobs tab
(A2:H9) and returns Min_Years.
```

4. Type Salary_Fit in S1, ask for the formula in S2, insert it and fill it down. S2 should say Within budget (S$6,200 against a Budget_Max of S$6,500).

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```
FORMULA:

Create a formula for S2 that returns "Within budget" if Expected_Salary in
K2 is less than or equal to Budget_Max in Q2, otherwise "Above budget".
```

5. Type Screen_Result in T1. Describe the whole rule in one prompt - Gemini nests the IFs for you. Insert and fill down. T2 (CAN-26481, score 65, 2 years) should say Possible. Screen_Result only flags candidates for a recruiter to review - a person still reads every application before anyone is rejected.

```
FORMULA:

Create a formula for T2: if Screening_Score in M2 is at least the pass mark
in Settings!B2 and Years_Experience in H2 is at least Min_Years in R2,
return "Strong"; otherwise, if M2 is at least Settings!B3, return
"Possible"; otherwise return "Not suitable".
```

6. Ask Gemini to explain the formula. Check its explanation against T2's actual values in M2, H2 and R2 - an explanation you can't match to the data is not an explanation.

```
ASK GEMINI:

Explain the formula in T2 step by step, and tell me what it returns for the
candidate in row 2 and why.
```

7. On the Checks tab, put labels in column A and ask Gemini for these two formulas one at a time, inserting each into column B. Expect 88 Strong in total and 14 Strong Data Analysts.

```
FORMULA:

1) Create a formula that counts how many cells in Applications!T2:T241 are
"Strong".

2) Create a formula that counts the rows on Applications where Position is
"Data Analyst" and Screen_Result is "Strong".
```

8. Paste the independent checks below them. Strong + Possible + Not suitable must add up to 240, and the Above budget count must match a COUNTIF of Salary_Fit. Then ask Gemini a what-if question and test its answer: change Settings!B2 to 75 - the Strong count should become 60 - then set it back to 70.

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```
CHECK:

Possible          =COUNTIF(Applications!T2:T241,"Possible")

Not suitable      =COUNTIF(Applications!T2:T241,"Not suitable")

All results       =COUNTA(Applications!T2:T241)

Above budget      =SUMPRODUCT(--(Applications!K2:K241>Applications!
Q2:Q241))


ASK GEMINI:

If I change the pass mark in Settings!B2 from 70 to 75, how many candidates
will be Strong?
```

9. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```

Sample prompts

Purpose	Prompt
Lookup	Create a formula that returns the Hiring_Manager for the Job_ID in E2 from the Jobs tab (A2:J9).
Date	Create a formula for the number of days between Applied_Date in D2 and the report date in Settings!B1.
Text	Create a formula that returns TRUE if Key_Skills in J2 contains "SQL".
Count	Create a formula that counts Above budget candidates for each Position.
Explain	Explain the difference between VLOOKUP and XLOOKUP, and when to use each.

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Test it

Check	Expected result
Fix	Q2 = S$6,500; no #REF! left in column Q.
Row 2	CAN-26481: Min_Years 2, Salary_Fit Within budget, Screen_Result Possible.
Screen_Result totals	Strong 88 + Possible 73 + Not suitable 79 = 240.
COUNTIFS	Strong Data Analysts = 14; Above budget = 62 (Within budget 178).
What-if	Pass mark 75 gives 60 Strong; set back to 70 gives 88.

Checkpoint and rejoin point

Topic 1 ends here. Lab 5 starts Topic 2 with its own tracker workbook.

Troubleshooting

If this happens	Fix
The Fix button does not appear.	Click the cell and hover over the red triangle. If there is still no Fix, use the ASK GEMINI prompt in Step 2 and type the corrected formula yourself.
Gemini inserted the formula in the wrong cell.	Undo, click the target cell first, then click Insert.
Gemini used XLOOKUP instead of VLOOKUP.	Both are correct. Ask 'Use VLOOKUP instead' if you want to practise VLOOKUP.
The formula works in row 2 but not lower down.	A range is not locked with $. Ask Gemini to 'make the Jobs range absolute' and fill down again.
Strong + Possible + Not suitable is not 240.	Some rows have no formula or an error. Filter column T for blanks or #N/A.

Challenge

Ask Gemini for one formula that returns the Strong count for every Position at once (hint: QUERY or a COUNTIFS against a list of positions). Check that it adds up to 88.

Reflection

Why is it better to keep the pass mark in Settings!B2 than to type 70 inside the formula?

Note: The lab folder labs/lab-04-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Screening_Formulas.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

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Topic 01 Recap - Mapped to Learning Outcomes

Use this checkpoint to explain what you can now do and identify any lab evidence you still need to repair.

Learning outcome	Evidence from this topic
LO1 · Side panel	Summaries, questions across chosen tabs, Insert, Retry, Undo and Clear history.
LO1 · Data quality	Duplicates, spelling variants, bad emails, text numbers and impossible values found and counted.
LO1 · Data prep	Gemini actions - find and replace, number format, insert/delete, freeze, sort, filter - with counts before and after.
LO1 · Formulas	VLOOKUP, IF, nested IF and COUNTIFS from plain English; explained, fixed and reconciled.

Topic 02 — Data Analytics and Visualization (Labs 5-8)

tables, trackers and dropdowns · conditional formatting · the AI function and Fill with Gemini · data analysis, insights and pivot tables · charts and graphs

Key concepts

- Structure from text — Gemini turns notes into a table you refine before you insert it.
- Rules that colour — Conditional formatting flags what matters; Settings lets you edit the rule.
- AI in a cell — =AI(prompt, range) extracts, categorises and summarises text row by row.
- Insights with evidence — Analysis steps show how Gemini calculated; you confirm with a formula.
- Pivot tables on request — Gemini builds a pivot table; you adjust it and check its totals.
- Inserted charts are copies — A chart Gemini inserts doesn't follow later changes - build linked ones too.

Concepts for Lab 5: Tables, Trackers and Dropdowns, Conditional Formatting with Gemini.

Tables, Trackers and Dropdowns

Gemini can create a table from scratch ('create an onboarding checklist') or from unstructured text such as meeting or interview notes, show it in the panel, let you refine it with follow-ups ('add an Outcome column') and insert it into the sheet. Dropdowns limit a column to valid values and checkboxes record yes/no states; Gemini adds both as actions.

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Trackers only work if everyone records things the same way. A table with dropdowns and checkboxes keeps data consistent from the start, so the analysis later is easy and trustworthy.

How it works

- Ask for the table with exact column names and allowed values.
- Refine with follow-ups, then Insert on a new tab.
- Add dropdowns and checkboxes with actions; test them.

Worked example

- 12 lines of interview notes become a 12-row Interview_Tracker.
- Status allows only Scheduled, Completed, No-show and Rescheduled.
- Reference_Checked is a checkbox column.

Decision guide

Use when	Avoid when
Information arrives as free text but must be tracked.	The notes contain sensitive details the tracker doesn't need.
Several people update the same sheet.	You insert before checking the draft against the notes.

Practitioner quality lens

- Columns: Named and ordered as requested.
- Values: Dropdowns restrict to valid options.
- Fidelity: Every note appears exactly once.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/186103
- https://support.google.com/docs/answer/14239833

Conditional Formatting with Gemini

Conditional formatting changes a cell's look when a rule is true: highlight rows over a threshold, values that exceed another column, duplicates or past dates, or shade values with a colour scale. Gemini creates the rule from a sentence ('Highlight rows where Screening_Score is 85 or more in light green'). The rules are visible and editable in Format > Conditional formatting or via Settings on the action card.

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Formatting makes exceptions jump out - the candidates above budget, the long notice periods - without changing the data. Because the rules are live, the colours update when the data changes.

How it works

- Say what to format, the condition and the colour; say 'whole row' if you mean it.
- Apply, then count the coloured cells with COUNTIF to confirm.
- Edit thresholds in the rule rather than adding a second rule.

Worked example

- Score 85+ rows green: 11 rows; edited to 80: 21 rows.
- Expected_Salary bold red when it is above Budget_Max: 12 cells.
- A red-to-green colour scale on Interview_Score.

Decision guide

Use when	Avoid when
You need exceptions to stand out for a reviewer.	Colour is the only signal - add a column for accessibility and filtering.
The rule is simple enough to state in one sentence.	Too many colours compete and nothing stands out.

Practitioner quality lens

- Rule: Visible in Format > Conditional formatting.
- Count: COUNTIF agrees with the coloured rows.
- Restraint: Few colours, each with a meaning.

Authoritative references

- https://support.google.com/docs/answer/78413
- https://support.google.com/docs/answer/14356410

Lab 5 · Tables, Trackers & Conditional Formatting — Build a Colour-Coded Interview Tracker from Unstructured Notes

Learning outcome: LO2: use Gemini to turn unstructured notes into a formatted table, build a tracker with dropdowns and checkboxes, and apply and edit conditional-formatting rules on HR data.

Goal: Turn a week of typed interview notes into a colour-coded tracker, and flag the strongest, most expensive and slowest-to-start candidates on the shortlist.

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The recruiters typed this week's interview notes as free text. You will ask Gemini to turn them into a structured table, refine it before inserting, and make it a working tracker with Status and Outcome dropdowns and a Reference_Checked checkbox. Conditional-formatting actions colour-code the tracker and flag the shortlist: top screening scores, salaries above budget and long notice periods. You will edit a rule with Settings, run a multi-step clean-up prompt, and create an onboarding checklist from scratch.

What you'll build

OrchidBay_Interview_Tracker with an Interview_Tracker tab (12 interviews, dropdowns, checkboxes, colour-coded Status), a flagged Shortlist tab and a new Onboarding_Checklist tab (Tools: Google Sheets · Gemini (create tables, Insert, dropdown, checkbox and conditional-formatting actions, Settings) · OrchidBay_Interview_Tracker.xlsx.)

Prerequisites

- Topic 1 completed: you can open the Gemini side panel, insert results and apply actions.
- This lab folder downloaded, including data/OrchidBay_Interview_Tracker.xlsx.

Step-by-step

1. Download OrchidBay_Interview_Tracker.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Interview_Tracker (Google Sheets copy)

Tabs      : About_This_Data | Interview_Notes (12 lines) | Shortlist (43
rows) | Prompt_Log
```

2. Ask Gemini to build a table from the notes. Read the draft in the panel and refine it with the follow-up before you insert it. Check that it has 12 rows and that every Status is one of the four options: Completed 5, Scheduled 3, Rescheduled 2, No-show 2. Click Insert, put it on a new tab and rename the tab Interview_Tracker.

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```
INSERT:

Using the notes on the Interview_Notes tab, create a table with the columns
Date, Time, Candidate, Candidate_ID, Position, Panel, Format and Status.
Status must be one of Scheduled, Completed, No-show or Rescheduled.



FOLLOW-UP:

Add two empty columns at the end: Outcome and Reference_Checked.
```

3. Make it a working tracker. Send these one at a time on the Interview_Tracker tab and click Apply on each action card. Test a dropdown by clicking a Status cell.

```
ACTION:

1) Make the Status column a dropdown with the options Scheduled, Completed,
No-show and Rescheduled.

2) Create a dropdown in the Outcome column with the options Pending,
Advance, Hold and Reject.

3) Insert checkboxes in the Reference_Checked column, unchecked by default.
```

4. Colour-code the Status column. When the action finishes, click Settings to see the rules Gemini created (the same rules are in Format > Conditional formatting). Change one Status to No-show and watch the colour change, then change it back.

```
ACTION:

Colour-code the Status column: Completed light green, Scheduled light blue,
Rescheduled amber and No-show light red.
```

5. Switch to the Shortlist tab and flag it one rule at a time. Expect 11 light-green rows (score 85 or more), 12 red salaries above budget, and a colour scale on Interview_Score (blank cells stay white).

```
ACTION:

1) On the Shortlist tab, highlight the rows where Screening_Score is 85 or
more in light green.

2) Make Expected_Salary bold red when it is greater than Budget_Max on the
same row.

3) Add a colour scale to Interview_Score from red for low values to green
for high values.
```

6. Edit a rule instead of starting again: open Format > Conditional formatting (or Settings on the first card), choose the green rule and change 85 to 80. Then ask Gemini to count the green rows and check its answer with the formula - both should be 21.

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```
ASK GEMINI:

On the Shortlist tab, how many rows have a Screening_Score of 80 or more?
Which positions are they for?



CHECK:

Score 80 or more   =COUNTIF(Shortlist!E2:E44,">=80")

Above budget       =SUMPRODUCT(--(Shortlist!G2:G44>Shortlist!H2:H44))
```

7. One multi-step prompt can do several actions. Check every part: Stage has a dropdown, Notice_End shows dates, Contacted has checkboxes, and 5 rows are light orange.

```
ACTION:

Clean up the Shortlist tracker: make the Stage column a dropdown with
Shortlisted, Interviewed and Offered; add a new column Notice_End at the
end with a formula for 30 September 2026 plus Notice_Weeks weeks, formatted
as a date; add a checkbox column called Contacted; and highlight in light
orange the rows where Notice_Weeks is more than 8.
```

8. Create a table from scratch - no data needed. Refine it, then Insert it on a new tab named Onboarding_Checklist.

```
INSERT:

Create an onboarding checklist tracker for a new hire in Singapore, with
the columns Task, Owner, Days_Before_Start, Status and Done. Include 10
typical tasks. Make Done a checkbox.



FOLLOW-UP:

Add 3 tasks for laptop, system access and security training, owned by IT.
```

9. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```


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Sample prompts

Purpose	Prompt
Table	Create a table of interview questions for a Data Analyst, with columns Question, Skill tested and What a good answer includes.
Dropdown	Convert the Recruiter column on the Shortlist tab to a dropdown with values from the same column.
Highlight	Highlight the Shortlist rows where Stage is Offered in light purple.
Freeze	Freeze the top row and the first two columns of the Shortlist tab.
Help	Teach me how to use conditional formatting with a custom formula.

Test it

Check	Expected result
Interview_Tracker	12 rows; Completed 5, Scheduled 3, Rescheduled 2, No-show 2.
Tracker controls	Status and Outcome dropdowns work; Reference_Checked has checkboxes; Status is colour-coded.
Shortlist flags	11 green rows at 85, then 21 after editing the rule to 80; 12 salaries in bold red; Interview_Score has a colour scale.
Multi-step	Stage dropdown, Notice_End dates, Contacted checkboxes; 5 light-orange rows.
Onboarding_Checklist	13 tasks (10 + 3 IT tasks) with a Done checkbox column.

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Checkpoint and rejoin point

Lab 6 opens a separate profiles workbook, so you can start it even if this tracker is unfinished.

Troubleshooting

If this happens	Fix
The table from the notes has wrong statuses.	Use Retry, or a follow-up: 'done or completed means Completed; moved means Rescheduled; did not turn up means No-show; confirmed or accepted means Scheduled.'
Insert put the table on top of the notes.	Undo. Insert it on a new tab, or select an empty cell on a new tab before clicking Insert.
The row highlight colours only one cell.	Undo and say 'highlight the whole row'. In Format > Conditional formatting the custom formula needs $ before the column, e.g. =$E2>=85, applied to A2:K44.
Dropdown or checkbox actions are not offered.	Use the menu: select the column > Insert > Dropdown (or Data > Data validation), and Insert > Checkbox.
No Settings button appears.	Settings appears only for actions with configurable settings. Use Format > Conditional formatting.

Challenge

Ask Gemini to add a Days_To_Start column and highlight anyone who cannot start before 1 December 2026.

Reflection

Which flag on the Shortlist would change a hiring manager's decision, and which is only 'nice to know'?

Note: The lab folder labs/lab-05-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Interview_Tracker.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Concepts for Lab 6: The AI Function (=AI), Fill with Gemini and AI Columns.

The AI Function (=AI)

The AI function puts Gemini in a cell: =AI("prompt", [range]) (or =Gemini(...)). It can generate text, summarise, categorise (including sentiment) and get real-time information from Google Search. You type the formula, select the cells and click Generate and insert; Refresh and insert regenerates. The range is the only data it sees - it doesn't read

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the whole sheet or Drive. Results are text, cannot be undone, are limited to 350 selected cells at a time and do not refresh automatically when inputs change.

It turns free text - candidate summaries, interview notes - into structured columns you can filter, count and chart. Because results are AI-generated, you measure their accuracy before you rely on them.

How it works

- Write a specific prompt with the output format ('Output only the number').
- Point the range at the cells it needs; fill down and Generate and insert.
- Compare a sample with known values; refresh after editing the prompt.

Worked example

- =AI("Extract the notice period in weeks ...", D2) turns 'one month's notice' into 4.
- =AI("Classify this interview feedback as Positive, Mixed or Negative ...", E2).
- A Checks tab counts how many AI values match the answer key.

Decision guide

Use when	Avoid when
Each row needs the same judgement on free text.	The output feeds a decision about a person without review.
You can check a sample against known answers.	You need numbers to calculate - convert and check first.

Practitioner quality lens

- Prompt: Specific, with the output format.
- Accuracy: Measured on a sample or answer key.
- Fairness: No personal characteristics in prompts or outputs.

Authoritative references

- https://support.google.com/docs/answer/15877199

Fill with Gemini and AI Columns

Fill with Gemini completes a column from the context of the table: type one or two examples and drag the fill handle, or select an empty range and choose Fill, optionally with a custom prompt. In a table, Insert AI column adds a column whose first row shows the AI function, ready to autofill. The side panel can also categorise a whole column at once and return a summary table.

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Examples are often easier than instructions: show Gemini one correct answer and it follows the pattern. Side-panel categorisation gives the big picture - themes and counts - that row-by-row formulas cannot.

How it works

- Type a correct example in the first cell.
- Select the range and use Fill with Gemini (or an AI column).
- Spot-check rows and ask the side panel for themes and counts.

Worked example

- Top_Skills: one example in I2, Gemini fills I3:I31.
- The side panel groups 30 feedback notes into themes.
- A bias check on the generated summaries.

Decision guide

Use when	Avoid when
The pattern is easy to show but hard to describe.	The example is wrong - errors spread down the column.
You need both per-row values and an overview.	The column will be used without spot-checks.

Practitioner quality lens

- Example: First value correct and typical.
- Coverage: Every row filled, none invented.
- Overview: Theme counts add up to the rows.

Authoritative references

- https://support.google.com/docs/answer/15877199
- https://blog.google/products-and-platforms/products/workspace/gemini-workspace-updates-march-2026/

Lab 6 · AI Function & Fill with Gemini — Extract and Categorise Candidate Text with the AI Function

Learning outcome: LO2: use the AI function (=AI) and Fill with Gemini to extract, categorise and summarise free-text HR data in cells, and measure the accuracy of the AI output against recorded values.

Goal: Turn 30 free-text candidate summaries and interview notes into structured columns - notice period, sentiment and a manager summary - and measure how accurate they are.

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Hiring managers want a quick read of 30 shortlisted candidates, but the details are buried in free text. You will type =AI formulas that extract the notice period in weeks, classify each interview feedback as Positive, Mixed or Negative, and write a one-line summary that leaves out personal characteristics. You will generate the outputs, compare them with an answer key using formulas, use Fill with Gemini for a skills column, and ask the side panel to group the feedback into themes.

What you'll build

OrchidBay_Candidate_Profiles with AI-filled Notice_Weeks_AI, Feedback_Sentiment_AI, Manager_Summary_AI and Top_Skills columns, a Checks tab measuring accuracy, and an inserted feedback-themes table (Tools: Google Sheets · AI function (=AI, Generate and insert) · Fill with Gemini · Gemini side panel · OrchidBay_Candidate_Profiles.xlsx.)

Prerequisites

- The AI function must be available on your account: type =AI(in a cell and check that it is recognised.
- This lab folder downloaded, including data/OrchidBay_Candidate_Profiles.xlsx.

Step-by-step

1. Download OrchidBay_Candidate_Profiles.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Candidate_Profiles (Google Sheets copy)

Tabs      : About_This_Data | Profiles (30 rows) | Answer_Key | Checks |
Prompt_Log

Empty     : F Notice_Weeks_AI | G Feedback_Sentiment_AI | H
Manager_Summary_AI
```

2. Type the formula into F2 exactly as shown, then select F2 and click Generate and insert. Check F2 against D2 (CAN-26495): the expected answer is 0. Then copy F2 down to F31 (drag the fill handle), select F2:F31 and click Generate and insert again. Up to 350 selected AI cells generate at once. The range argument (D2) is the only data the function sees.

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```
AI FUNCTION:

=AI("Extract the notice period in weeks as a whole number. Output only the
number. Use 0 for immediately, 4 for one month, 8 for two months and 12 for
three months.", D2)
```

3. Categorise the feedback the same way in G2, then fill and generate G2:G31. Row 2 should be Mixed: 'Solid background, but answers on B2B Sales lacked depth. Consider a second interview.'

```
AI FUNCTION:

=AI("Classify this interview feedback as Positive, Mixed or Negative.
Output one word only.", E2)
```

4. Write a manager summary in H2 from three cells at once. Generate only H2:H6 - summaries use more of your AI generation limit. Read each one: does it stick to the evidence?

```
AI FUNCTION:

=AI("Write a one-sentence summary of this candidate for the hiring manager,
using only the summary and feedback. Do not mention age, gender, race,
religion or nationality.", C2:E2)
```

5. Measure accuracy. Paste the check formulas into Checks column B (labels in column A). Answer_Key holds the recorded values: Positive 9, Mixed 16, Negative 5. Note how many of the 30 match. AI output is text and cannot be undone - to change an answer, edit the prompt and click Refresh and insert.

```
CHECK:

Notice matches      =SUMPRODUCT(--(TO_TEXT(Profiles!
F2:F31)=TO_TEXT(Answer_Key!B2:B31)))

Sentiment matches   =SUMPRODUCT(--(TRIM(Profiles!G2:G31)=Answer_Key!
C2:C31))

Positive (AI)       =COUNTIF(Profiles!G2:G31,"Positive")

Mixed (AI)          =COUNTIF(Profiles!G2:G31,"Mixed")
```

6. Investigate any mismatch with the side panel. Now that your answers are generated, it is fine for Gemini to read the answer key - it was kept out of the =AI prompts so it could not influence them.

```
ASK GEMINI:

Compare Profiles column G with Answer_Key column C. Which candidates don't
match, and what in the Interview_Feedback text could have caused each
difference?
```

7. Fill with Gemini: type Top_Skills in I1 and, in I2, the skills from D2 yourself (e.g. 'CRM, Negotiation, Stakeholder Management'). Select I2:I31 and choose Fill (or drag

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the fill handle down and pick Fill with Gemini when it is offered). Gemini follows your example for every row. If Fill is not offered on your account, use the AI function instead.

```
AI FUNCTION:

=AI("List the skills mentioned in this summary, comma separated, in the
same order.", D2)
```

8. Use the side panel to categorise the whole column at once, and insert the result on a new tab named Feedback_Themes. Then check for bias in the generated summaries.

```
INSERT:

Group the Interview_Feedback on the Profiles tab into themes and count how
many candidates fall into each theme. Show the result as a table with one
example per theme.


ASK GEMINI:

Do any of the summaries in column H mention personal characteristics or
anything that should not be used in a hiring decision?
```

9. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```

Sample prompts

Purpose	Prompt
AI function	=AI("Does this candidate mention SQL? Answer Yes or No.", D2)
AI function	=AI("Write a polite two-sentence email inviting this candidate to a second interview.", B2:C2)
Categorise	=AI("Classify the seniority of this candidate as Junior, Mid or Senior based on years of experience.", D2)
Side panel	Which five candidates have the most positive feedback? Show their names and positions.
Side panel	Summarise the main concerns interviewers raised, with the number of candidates for each.

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Test it

Check	Expected result
Notice_Weeks_AI	30 numbers; row 2 = 0; Notice matches close to 30 (recorded: 8 candidates with 8+ weeks).
Feedback_Sentiment_AI	Recorded split Positive 9, Mixed 16, Negative 5; your Sentiment matches count shows how many agree.
Manager_Summary_AI	H2:H6 filled with one sentence each and no personal characteristics.
Top_Skills	I2:I31 filled by Fill with Gemini (or =AI), following your example in I2.
Feedback_Themes	An inserted themes table whose counts add up to 30.

Checkpoint and rejoin point

Lab 7 opens the 2026 year-to-date workbook - 720 applications including the Q3 drive.

Troubleshooting

If this happens	Fix
'AI function not available'.	Your plan, admin settings or language setting does not include it. Use the side panel instead: 'For each row, extract the notice period in weeks from Candidate_Summary' and insert the result.
The cell shows the formula but no answer.	Select the cell(s) and click Generate and insert - AI cells do not generate on their own.
You can't generate any more.	You reached a generation limit. Generate fewer cells (e.g. 10 rows) or wait and try later.
Answers come back as '4 weeks' instead of 4.	Make the prompt stricter ('Output only the number'), then Refresh and insert.
The AI output changed after you edited D2.	AI results do not update on their own. Select the cell and click Refresh and insert.

Challenge

Write one =AI formula that returns 'Ready now', 'Within a month' or 'Later' from the summary. Check it against Notice_Weeks_AI with an IFS formula.

Reflection

For which kinds of HR text would you never let =AI output go into a record without a person reading it?

Note: The lab folder labs/lab-06-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Candidate_Profiles.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

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Concepts for Lab 7: Data Analysis and Insights with Gemini, Pivot Tables with Gemini.

Data Analysis and Insights with Gemini

Gemini answers questions about the data in your sheet: identify trends, compare groups, calculate rates and averages, suggest analyses ('What analysis can you help me perform with this sheet?') and even discuss regression and prediction. Analysis steps shows how it calculated (and Show code, where offered), and Export to Docs sends the response to a new Google Doc. With 'Use Google Search' in the prompt it can bring in web information.

Managers want answers, not tables. Gemini gets you to a first answer in seconds; Analysis steps and a formula check make it defensible. Web answers are external evidence and need their sources checked.

How it works

- Ask a specific question with the measure, group and period.
- Open Analysis steps; confirm the key number with COUNTIFS or AVERAGEIFS.
- Ask for insights with numbers and actions, then Export to Docs.

Worked example

- Applications peaked in June 2026 (98).
- Referral has the best hire rate (12.8%); LinkedIn brings the most hires (11).
- HR Executive takes longest to fill (39.3 days).

Decision guide

Use when	Avoid when
You need a quick, explainable answer from a dataset.	A small group drives the result (e.g. 78 referrals) and nobody checks it.
You will check the key numbers before sharing.	Web results without sources.

Practitioner quality lens

- Definition: Measures defined (what is a hire?).
- Traceability: Analysis steps read; formula check done.
- Context: Group sizes stated with every rate.

Authoritative references

- https://support.google.com/docs/answer/14356410

Pivot Tables with Gemini

A pivot table summarises a large table by grouping rows and aggregating values - counts, sums, averages - with fields in rows, columns and filters. Gemini creates one

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from a sentence ('Create a pivot table that shows the count of Candidate_ID with Source in rows and Stage in columns'), usually on a new sheet, and Settings opens the pivot table editor to adjust it.

Pivot tables answer 'how many / how much by group' questions and stay linked to the data, so they refresh when it changes. Combined with Gemini you get the structure without hunting through the editor.

How it works

- Name the value, the aggregation and the row/column fields.
- Apply, then adjust in Settings (filters, sorting, totals).
- Check one cell and the grand total with COUNTIFS.

Worked example

- Source x Stage: LinkedIn Hired 11, Rejected 147; grand total 720.
- Average Days_To_Hire by Position, sorted high to low.
- Both checked with COUNTIFS / AVERAGEIFS.

Decision guide

Use when	Avoid when
You need a grouped summary that updates with the data.	The source data still has duplicates or spelling variants.
The same breakdown is needed regularly.	A single number would do - use COUNTIFS.

Practitioner quality lens

- Fields: Rows, columns and values as intended.
- Total: Grand total equals the row count.
- Check: One cell confirmed by formula.

Authoritative references

- https://support.google.com/docs/answer/1272900
- https://support.google.com/docs/answer/14356410

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Lab 7 · Data Analysis & Insights — Generate Hiring Insights and Pivot Tables with Gemini

Learning outcome: LO2: use Gemini to analyse an HR dataset - trends, comparisons and rates - inspect its analysis steps, create and adjust pivot tables, verify every number with formulas and export the insights to Google Docs.

Goal: Answer the Head of Talent's questions about 2026 hiring - trend, best source, slowest role, cost per hire - with numbers you have checked.

Orchid Bay's Head of Talent wants to know how 2026 hiring is going. Using all 720 applications from January to September, you will ask Gemini what analysis is possible, find the monthly trend, compare sources by hire rate and cost per hire, and find which role takes longest to fill. You will open Analysis steps to see how Gemini calculated, create pivot tables with Gemini actions and adjust them with Settings, check every figure with COUNTIFS or AVERAGEIFS, and export five insights to a Google Doc.

What you'll build

OrchidBay_Hiring_2026_YTD with two pivot tables (source by stage, days to hire by position), a Checks area of formulas, and a Google Doc of insights exported from Gemini (Tools: Google Sheets · Gemini data analysis (Analysis steps, Export to Docs) · pivot-table actions · OrchidBay_Hiring_2026_YTD.xlsx.)

Prerequisites

- Labs 1-6, or comfort with Ask Gemini, Insert and actions.
- This lab folder downloaded, including data/OrchidBay_Hiring_2026_YTD.xlsx.

Step-by-step

1. Download OrchidBay_Hiring_2026_YTD.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Hiring_2026_YTD (Google Sheets copy)

Tabs      : About_This_Data | Applications (720 rows) | Jobs | Source_Costs
| Prompt_Log

Period    : applications 2026-01-01 to 2026-09-26
```

2. Start broad. Gemini suggests analyses that fit the columns it finds - note two you want to try.

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```
ASK GEMINI:

What analysis can you help me perform with this sheet?
```

3. Ask for the trend. Under the answer, click Analysis steps (and Show code if offered) to see how Gemini grouped and counted. Applications peaked in June (98); February was lowest (62).

```
ASK GEMINI:

Identify trends in the number of applications by Month. Which month had the
most applications, and which had the fewest?
```

4. Compare sources. Gemini should find that Referral has the best hire rate (10 of 78, 12.8%) while LinkedIn brings the most hires (11 of 184).

```
ASK GEMINI:

For each Source, show the applications, hires (Stage is Hired) and hire
rate. Which source has the best hire rate, and which brings in the most
hires?
```

5. Create a pivot table with a Gemini action. When it finishes, click Settings to open the pivot table editor - try adding a filter. LinkedIn should show Hired 11 and Rejected 147; the grand total is 720. Then paste the checks into empty cells beside the pivot table.

```
ACTION:

Create a pivot table on a new sheet that shows the count of Candidate_ID
with Source in rows and Stage in columns.


CHECK:

LinkedIn hired      =COUNTIFS(Applications!G2:G721,"LinkedIn",Applications!
M2:M721,"Hired")

Referral hired      =COUNTIFS(Applications!G2:G721,"Referral",Applications!
M2:M721,"Hired")

All applications    =COUNTA(Applications!A2:A721)
```

6. Which role is slowest to fill? Gemini should name HR Executive (39.3 days on average; 31.1 days across all 41 hires). Confirm with a second pivot table and an AVERAGEIFS check.

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```
ASK GEMINI:

What is the average Days_To_Hire for each Position? Which role takes the
longest to fill?


ACTION:

Create a pivot table that shows the average Days_To_Hire by Position,
sorted from highest to lowest.


CHECK:

HR Executive avg   =AVERAGEIFS(Applications!P2:P721,Applications!
F2:F721,"HR Executive",Applications!M2:M721,"Hired")
```

7. Bring in costs from a second tab. Cost per hire = applications x cost per application / hires. Company Website should be cheapest (S$125) and Career Fair dearest (S$1,540).

```
ASK GEMINI:

Using the Applications and Source_Costs tabs, calculate the total cost and
the cost per hire for each Source. Which source gives the cheapest hires?
```

8. Ask for insights for the Head of Talent. Check each number against your pivot tables and checks, then click Export to Docs and name the document 'Orchid Bay 2026 Hiring Insights'. Optional: ask the web question - Gemini only searches the web when you ask it to, and you must check its sources.

```
ASK GEMINI:

Write five insights for the Head of Talent from this data. Give the number
behind each insight and one recommended action.


ASK GEMINI:

Using Google Search, what is a typical time to hire for professional roles
in Singapore? List your sources.
```

9. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

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```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```

Sample prompts

Purpose	Prompt
Trend	How did the number of hires change from Q1 to Q2 to Q3 2026?
Compare	Compare the average Screening_Score of hired and rejected candidates.
Pivot	Create a pivot table that shows the number of hires by Recruiter and Month.
Regression	How can I show regression and prediction of monthly applications?
Explain	Explain this analysis like I'm a hiring manager with no data background.

Test it

Check	Expected result
Trend	Peak June 2026 (98 applications); lowest February (62).
Sources	Best hire rate Referral 12.8%; most hires LinkedIn (11); 41 hires in total.
Pivot 1	Source x Stage; LinkedIn Hired 11, Rejected 147; grand total 720.
Pivot 2	HR Executive slowest at 39.3 days; AVERAGEIFS matches.
Cost per hire	Cheapest Company Website S$125; dearest Career Fair S$1,540; insights exported to a Google Doc.

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Checkpoint and rejoin point

Lab 8 charts the same data from its own copy of the workbook.

Troubleshooting

If this happens	Fix
No Analysis steps button.	It appears only for answers Gemini calculated from the sheet. Ask for a calculation, e.g. 'count by Month'.
Gemini counts 721 applications.	It included the header row. Ask it to count data rows only, or check with =COUNTA(Applications!A2:A721).
Hire rate differs from yours.	Ask how it defined a hire. It must be Stage = Hired, divided by all applications from that source.
No action card for the pivot table.	Build it from the menu: select the data > Insert > Pivot table > New sheet, then add Source to Rows, Stage to Columns and COUNTA of Candidate_ID to Values.
Export to Docs is missing.	It is shown for analysis responses. Use Copy, then paste into a new Google Doc.
The web answer has no sources.	Do not use it. Ask again with 'list your sources', or use Retry with Google Search.

Challenge

Ask Gemini whether candidates with a higher Screening_Score are more likely to be hired, and check its claim with two AVERAGEIFS.

Reflection

Referral has the best hire rate but only 78 applications. What would you check before telling the business to 'do more' of it?

Note: The lab folder labs/lab-07-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Hiring_2026_YTD.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Concepts for Lab 8: Charts and Graphs with Gemini, Inserted vs Linked Charts.

Charts and Graphs with Gemini

Gemini generates charts from a description - 'Create a chart with Month on the x-axis and applications on the y-axis', 'Make a bar chart of hires by Source'. You can Preview the chart, Copy it, check Analysis steps, Export to Docs, and Insert it. Inserting adds a new tab with the chart and its underlying data; the chart is linked to that new tab's data and fully editable.

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The right chart makes a finding obvious in seconds. Asking for the chart type, axes and sorting gets a usable first draft; you then make the title state the finding and add alt text for accessibility.

How it works

- Describe the chart type, axes, measure and sort order.
- Preview, then Insert; read the new-tab message.
- Rewrite the title as the finding; add alt text.

Worked example

- Line: applications by month - 'Applications peaked in June'.
- Sorted bar: hires by source - LinkedIn first.
- Column: the hiring funnel 720 / 61 / 47 / 41.

Decision guide

Use when	Avoid when
You need a quick, editable chart of a clear question.	The chart must follow new data - see Inserted vs Linked Charts.
The chart is a one-off snapshot for a report.	The chart type hides the story (e.g. a pie with many slices).

Practitioner quality lens

- Fit: Chart type matches the question.
- Title: States the finding.
- Access: Alt text added.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/63824

Inserted vs Linked Charts

A chart inserted from the Gemini panel is built on a copy of the data placed on its own new tab. Google's guidance is explicit: the generated chart doesn't link or respond to changes in the original data set used to generate it. A chart you build with Insert > Chart on a live range - or on a pivot table or formula table - updates whenever that range changes.

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Reports and dashboards must stay true when the data changes. Knowing which charts are snapshots and which are live prevents a dashboard from quietly showing last week's numbers.

How it works

- Use Gemini-inserted charts for snapshots and drafts.
- For live charts, build a pivot table or formula table, then Insert > Chart on it.
- Test: change one source value and see which charts move.

Worked example

- Changing one Stage to Hired: the inserted chart stays the same.
- The chart on the Month pivot table updates.
- Dashboards in Topic 3 use only linked charts.

Decision guide

Use when	Avoid when
Snapshot: a one-off chart in a report or email.	Using an inserted chart on a dashboard that must refresh.
Live: dashboards and recurring reports.	Forgetting to undo test edits to the source data.

Practitioner quality lens

- Purpose: Snapshot or live - decided on purpose.
- Test: One value changed and checked.
- Reset: Test edits undone.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/63824

Lab 8 · Charts & Visualisation — Build Charts That Tell the Hiring Story

Learning outcome: LO2: generate line, bar, column and combination charts with Gemini, preview and insert them, understand which charts update with the data, build linked charts and write titles that state the finding.

Goal: Turn the 2026 hiring data into four clear charts - trend, sources, funnel, cost per hire - and know which ones update when the data changes.

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Now you show the findings from Lab 7. You will ask Gemini to create charts - a monthly trend line, hires by source and the hiring funnel - preview them and insert them. Each inserted chart lands on a new tab with its own copy of the data, so it does not update when Applications changes. You will prove that, then build a linked chart from a pivot table that does update. A cost-per-hire chart combines two tabs, and you will rewrite every title to state its finding and add alt text.

What you'll build

OrchidBay_Hiring_Charts with four inserted Gemini charts, one linked chart built from a pivot table, finding-based titles and alt text (Tools: Google Sheets · Gemini charts (Preview, Insert, Copy chart) · pivot table · Chart editor · OrchidBay_Hiring_Charts.xlsx.)

Prerequisites

- Lab 7 completed, or its Test It values to hand.
- This lab folder downloaded, including data/OrchidBay_Hiring_Charts.xlsx.

Step-by-step

1. Download OrchidBay_Hiring_Charts.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Hiring_Charts (Google Sheets copy)

Tabs      : About_This_Data | Applications (720 rows) | Jobs | Source_Costs
| Prompt_Log
```

2. Ask for a trend chart. Click Preview to see it, then Insert. Gemini adds a NEW tab with the chart and the data behind it - read the message at the bottom left. The line should peak in June (98).

```
INSERT:

Create a line chart with Month on the x-axis and the number of applications
on the y-axis.
```

3. Ask for two more charts and insert each. Hires by source, sorted: LinkedIn 11, Referral 10, JobStreet 8 ... Funnel: 720, 61, 47, 41.

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```
INSERT:

Make a bar chart of hires (Stage is Hired) by Source, sorted from most to
fewest.



INSERT:

Create a column chart of the hiring funnel: Applications (all rows),
Interviewed (rows with an Interview_Score), Offered (Stage is Offered or
Hired) and Hired.
```

4. Test whether the charts update. On Applications, change the Stage of one LinkedIn row from Rejected to Hired. Look at the hires-by-source chart: it does NOT change, because it is linked to the copy of the data on its own tab. Ask Gemini why, then undo your edit (Ctrl + Z / Cmd + Z).

```
ASK GEMINI:

Will the chart you inserted update if I change the Applications tab? How do
I make a chart that does?
```

5. Build a linked chart. Ask Gemini for a pivot table, click Apply, then select the pivot table and choose Insert > Chart and pick a column chart. Repeat the Stage edit from Step 4: the Hired count for that month goes up by one in the pivot table and in this chart. Undo the edit again.

```
ACTION:

Create a pivot table on a new sheet that counts Candidate_ID with Month in
rows and Stage in columns.
```

6. Combine two tabs in one chart. The lowest bar should be (S$125, Company Website) and Career Fair the highest (S$1,540).

```
INSERT:

Using the Applications and Source_Costs tabs, create a column chart of cost
per hire by Source, where cost per hire is applications x
Cost_Per_Application divided by hires.
```

7. Make every chart say something. Ask for finding-based titles, then double-click each chart > Customize > Chart & axis titles to apply them. Right-click each chart > Alt text and add one sentence. Optional: More options > Copy chart in the panel, and paste it into a Google Doc or Slides.

```
ASK GEMINI:

Suggest a title for each chart in this spreadsheet that states the main
finding, not just the measure.

For example: 'Applications peaked in June'.
```


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8. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```

Sample prompts

Purpose	Prompt
Pie	Create a pie chart of applications by Source.
Stacked	Create a stacked column chart of applications by Month, split by Stage.
Combo	Create a chart with applications as columns and hires as a line, by Month.
Help	What chart type is best to show the share of each source over time?
Graph	Create a graph that shows the percentage of applications that came from each Position.

Test it

Check	Expected result
Trend chart	Line chart on its own tab; peak June 98.
Source and funnel	Bars sorted; LinkedIn 11 first. Funnel 720 / 61 / 47 / 41.
Static vs linked	Inserted chart did not change after the Stage edit; the Month x Stage pivot chart did.
Cost per hire	Company Website S$125 lowest; Career Fair S$1,540 highest.
Titles and alt text	Every chart title states a finding; every chart has alt text.

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Checkpoint and rejoin point

Topic 2 ends here. Lab 9 starts the interactive dashboard in its own workbook.

Troubleshooting

If this happens	Fix
No Preview or Insert button.	Ask again with the word 'chart' and the chart type. Make sure you are in the Google Sheets copy.
The chart counts the header as a month.	Ask Gemini to use data rows only, or edit the data range in the Chart editor.
Gemini put the chart on a new tab, but you wanted it on Applications.	That is how Insert works. Cut the chart (three dots > Cut) and paste it on another tab, or build it with Insert > Chart.
You can't find Copy chart.	In the Gemini panel, open More options (three dots) on the chart response.

Challenge

Build a combo chart from the Lab 8 pivot table that shows applications (columns) and hire rate (line on a second axis).

Reflection

When is a static chart fine, and when must a chart be linked to live data?

Note: The lab folder labs/lab-08-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Hiring_Charts.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Topic 02 Recap - Mapped to Learning Outcomes

Use this checkpoint to explain what you can now do and identify any lab evidence you still need to repair.

Learning outcome	Evidence from this topic
LO2 · Tables and trackers	Notes turned into a colour-coded tracker with dropdowns, checkboxes and editable rules.
LO2 · AI function	=AI extracts, categorises and summarises text; accuracy measured against an answer key.
LO2 · Analysis	Trends, rates and cost per hire with Analysis steps; pivot tables checked with COUNTIFS.
LO2 · Charts	Inserted (static) and linked charts; titles that state the finding; alt text.

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Topic 03 — Interactive Dashboard (Labs 9-12)

dropdown selectors and KPI formulas · linked charts, alerts and protection · Gems for repeatable screening · optimisation, reporting and responsible sharing

Key concepts

- Selectors drive everything — Dropdowns feed COUNTIFS; 'All' becomes the wildcard *.
- KPIs are formulas — Never type a number Gemini told you - the dashboard must recalculate.
- Charts on live ranges — Insert > Chart on the dashboard tables, so charts follow the dropdowns.
- Gems for repeat work — Saved instructions + knowledge files = the same screening every time.
- Optimise, then decide — Gemini's solver finds the best combination; a person makes the call.
- Share the minimum — Only the fields a viewer needs, view-only, after the external-sharing warning.

Concepts for Lab 9: What Makes a Dashboard Interactive, KPI Formulas Driven by Dropdowns.

What Makes a Dashboard Interactive

An interactive dashboard is one page of KPIs, tables and charts that viewers can re-cut themselves. In Google Sheets the interactivity comes from selector cells - dropdowns - whose values feed the formulas behind every KPI and table, so changing a dropdown recalculates the whole page.

One dashboard answers many questions ('how are we doing for Software Engineers from LinkedIn?') without building a new report each time. Because everything is formula-driven, it stays correct when data is added.

How it works

- Plan the questions, the KPIs and the selectors (e.g. Position, Source).
- Build a selector with a dropdown whose list includes 'All'.
- Make every KPI and table read the selectors.

Worked example

- C3 Position and C4 Source drive seven KPIs.
- Funnel, monthly and source tables use the same filters.
- Software Engineer: 104 applications, 6 hires.

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Decision guide

Use when	Avoid when
The same questions are asked for different groups.	A one-off answer - a pivot table or Gemini question is faster.
Data is refreshed regularly.	Groups are so small that individuals could be identified.

Practitioner quality lens

- Questions: Each KPI answers a stated question.
- Drivers: Every number reads the selectors.
- Tests: At least three views checked.

Authoritative references

- https://support.google.com/docs/answer/186103
- https://support.google.com/docs/answer/3256550

KPI Formulas Driven by Dropdowns

Each KPI is a formula that filters by the selectors. COUNTIFS counts rows that meet every condition; AVERAGEIFS averages a column for matching rows. The pattern IF($C$3="All","*",$C$3) turns 'All' into the wildcard *, which matches any value. IFERROR shows 0 or '-' instead of #DIV/0! when nothing matches.

Formula KPIs recalculate instantly and can be audited cell by cell. Gemini writes them from a description, but you must test edge cases - 'All', a single group, and a combination with no hires.

How it works

- Describe the count, the filters and what 'All' means.
- Insert and lock references with $ before filling down.
- Test All/All, one group, and an empty combination.

Worked example

- B7 =COUNTIFS(...F...,IF($C$3="All","*",$C$3),...G...,IF($C$4="All","*",$C$4)) = 720.
- Hire rate =IFERROR(E7/B7,0) = 5.7%.
- Data Analyst + LinkedIn: 30 applications, 0 hires, '-' days.

Decision guide

Use when	Avoid when
Counts, rates and averages filtered by selectors.	A KPI needs logic you can't express or check in a formula.
The same formula must work for every group.	Typed numbers instead of formulas.

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Practitioner quality lens

- Wildcard: 'All' handled with *.
- Errors: IFERROR on every ratio.
- Edges: Empty combinations tested.

Authoritative references

- https://support.google.com/docs/answer/3256550
- https://support.google.com/docs/answer/3093364

Lab 9 · Dashboard KPIs & Selectors — Build the Dashboard's Selectors and KPI Formulas

Learning outcome: LO3: use Gemini to add dropdown selectors and generate the COUNTIFS and AVERAGEIFS formulas that drive a dashboard's KPIs and summary tables, and test that every KPI responds correctly to the selectors.

Goal: Build the formula layer of an interactive hiring dashboard: two dropdowns (Position, Source) that drive seven KPIs and three summary tables.

The Head of Talent wants one page that answers 'how is hiring going?' for any role and any source. You will turn two cells into dropdown selectors, then ask Gemini for the formulas: COUNTIFS that treat 'All' as 'every value', a hire rate, AVERAGEIFS for screening score and days to hire, and the funnel, monthly and source tables that the charts in Lab 10 will use. Every KPI is a formula, never a typed number, so it changes when the selectors change - and you test that it does.

What you'll build

OrchidBay_Dashboard_Build with Position and Source dropdowns in C3:C4, a KPI row (B7:H7) and funnel, monthly and source tables that all respond to the dropdowns (Tools: Google Sheets · Gemini dropdown actions and formula generation (COUNTIFS, AVERAGEIFS, IFERROR) · OrchidBay_Dashboard_Build.xlsx.)

Prerequisites

- Lab 4 (formulas) and Lab 7 (analysis) completed, or comfort with COUNTIFS.
- This lab folder downloaded, including data/OrchidBay_Dashboard_Build.xlsx.

Step-by-step

1. Download OrchidBay_Dashboard_Build.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx

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tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Dashboard_Build (Google Sheets copy)

Tabs      : About_This_Data | Dashboard | Applications (720 rows) | Jobs |
Lists | Prompt_Log

Dashboard : C3 Position, C4 Source, KPI labels B6:H6, tables at B10, E10
and I10
```

2. Turn the two yellow cells into selectors, one action at a time. Then set both to All.

```
ACTION:

1) On the Dashboard tab, create a dropdown in C3 with the values from
Lists!A2:A10.

2) On the Dashboard tab, create a dropdown in C4 with the values from
Lists!B2:B9.
```

3. Click B7 and ask for the core formula. The trick is that 'All' must mean 'any value' - Gemini usually replaces it with the wildcard "*". Insert it: with All / All, B7 must show 720.

```
FORMULA:

Create a formula for Dashboard!B7 that counts the rows on Applications
(rows 2-721) where Position in column F equals C3 and Source in column G
equals C4. If C3 or C4 is "All", count every value for that column.
```

4. Ask for the next three KPIs one at a time, clicking the target cell before each Insert. Expect Interviews 61, Offers 47 and Hires 41.

```
FORMULA:

1) Create a formula for C7: the same count as B7, but only rows where
Interview_Score (column L) is not blank.

2) Create a formula for D7: the same count as B7, but only rows where Stage
(column M) is Offered or Hired.

3) Create a formula for E7: the same count as B7, but only rows where Stage
(column M) is Hired.
```

5. Now the rate and averages. Format F7 as a percentage and G7:H7 with one decimal place. With All / All: hire rate 5.7%, average screening score 65.5, average days to hire 31.1.

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```
FORMULA:

1) Create a formula for F7 that divides Hires in E7 by Applications in B7,
showing 0 if B7 is 0.

2) Create a formula for G7: the average Screening_Score (column K) for the
rows that match C3 and C4.

3) Create a formula for H7: the average Days_To_Hire (column P) for Hired
rows that match C3 and C4, showing "-" if there are none.
```

6. Fill the tables the charts will use. The funnel just points at the KPIs. For the monthly table, insert the formulas in F11 and G11, then fill down to row 19. For the source table, insert J11 and fill down to J17. With All / All, June (F16) = 98 and LinkedIn hires (J11) = 11.

```
ACTION:

Fill Dashboard!C11:C14 with the formulas =B7, =C7, =D7 and =E7.


FORMULA:

1) Create a formula for F11 that counts applications where Month
(Applications column D) equals E11, using the same Position and Source
filters as B7.

2) Create a formula for G11: the same as F11 but only Hired rows.

3) Create a formula for J11 that counts Hired rows where Source equals I11
and Position matches C3 (All means any).
```

7. Test the selectors. Set C3 to Software Engineer (C4 All): Applications 104, Hires 6, hire rate 5.8%. Set C3 to Data Analyst and C4 to LinkedIn: 30 applications and 0 hires, and H7 shows '-' instead of an error. Ask Gemini to check one view, and confirm with the formula.

```
ASK GEMINI:

On the Dashboard tab, Position is set in C3. Check the KPI row B7:H7
against the Applications tab - are the counts and averages correct for that
position?


CHECK:

SE hires   =COUNTIFS(Applications!F2:F721,"Software Engineer",Applications!
M2:M721,"Hired")
```

8. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown.

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Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```

Sample prompts

Purpose	Prompt
KPI	Create a formula for the offer acceptance rate: Hired divided by Offered plus Hired, for the selected Position and Source.
KPI	Create a formula that returns the most common Source for the selected Position.
Selector	Create a dropdown in C5 with the values 2026-01 to 2026-09.
Explain	Explain the formula in B7 to someone who has never used COUNTIFS.
Check	Which formulas on the Dashboard tab would break if a new Position were added?

Test it

Check	Expected result
Selectors	C3 lists All + 8 positions; C4 lists All + 7 sources.
All / All	B7:H7 = 720, 61, 47, 41, 5.7%, 65.5, 31.1.
Tables	Funnel 720/61/47/41; June applications 98; LinkedIn hires 11.
Software Engineer	104 applications, 6 hires, 5.8%; the COUNTIFS check shows 6.
Data Analyst + LinkedIn	30 applications, 0 hires; H7 shows '-' (no error).

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Checkpoint and rejoin point

Lab 10 provides these finished KPI formulas in its own workbook, so you can continue even if a formula here is not working.

Troubleshooting

If this happens	Fix
B7 shows 0 with All / All.	The formula is comparing to the text 'All'. Ask Gemini: 'when C3 is All, use the wildcard "*"'.
Numbers are right for All but wrong for one position.	Check the selector spelling matches Applications exactly - pick from the dropdown, do not type.
No dropdown action, or Gemini's formula doesn't insert.	Dropdown: select C3 > Insert > Dropdown > 'Dropdown (from a range)' > Lists!A2:A10. Formulas: copy the formula shown under 'The formula Gemini should write' on the prompts page and adapt the condition.
#DIV/0! in F7 or H7.	Wrap the formula in IFERROR - ask Gemini to 'show 0 (or -) instead of an error'.
Filling down F11 changes C3 to C4.	The selector references need $ ($C$3, $C$4). Ask Gemini to make them absolute.

Challenge

Add an Offer acceptance KPI (Hired / (Offered + Hired)) in I7 and test it for Customer Success Executive.

Reflection

Why must every KPI on a dashboard be a formula rather than a number Gemini told you?

Note: The lab folder labs/lab-09-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Dashboard_Build.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Concepts for Lab 10: Dashboard Design: Charts, Alerts and Protection.

Dashboard Design: Charts, Alerts and Protection

A finished dashboard puts selectors at the top, KPIs in a row, and three or four charts below, each linked to a dashboard table so it follows the dropdowns. A metric switch (a dropdown plus an IF) lets one chart show different measures. Conditional formatting turns KPI cells into alerts. Hiding gridlines and helper tabs and protecting everything except the selectors makes it safe to share.

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A dashboard is used by people who didn't build it. Clear layout, alerts that draw the eye and protection against accidental edits make it trustworthy in other hands.

How it works

- Build charts with Insert > Chart on the dashboard tables.
- Add a metric switch and KPI alerts with Gemini actions.
- Hide gridlines and helper tabs; protect all but the selectors; test three views.

Worked example

- Customer Success Executive: 103 applications, 7 hires - all charts redraw.
- Hire rate below 5% turns red; days to hire above 35 turns amber.
- Viewers can change C3:C5 but cannot edit formulas.

Decision guide

Use when	Avoid when
Others will use the dashboard regularly.	Gemini-inserted (static) charts on a live dashboard.
A few KPIs need attention when they cross a threshold.	Alerts on every cell - nothing stands out.

Practitioner quality lens

- Layout: Selectors > KPIs > charts.
- Live: Every chart follows the dropdowns.
- Safe: Formulas protected; helpers hidden.

Authoritative references

- https://support.google.com/docs/answer/63824
- https://support.google.com/docs/answer/78413
- https://support.google.com/docs/answer/1218656

Lab 10 · Interactive Dashboard — Make the Hiring Dashboard Interactive and Visual

Learning outcome: LO3: turn the KPI formulas into an interactive, visual dashboard - linked charts, a metric switch, conditional-formatting alerts, protection and a Gemini summary exported to Google Docs.

Goal: Finish a one-page hiring dashboard whose charts, alerts and KPIs all respond to the dropdowns - and prove it with three test views.

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The KPI formulas from Lab 9 are ready: dropdowns in C3:C4 drive the KPIs and three tables. You will add charts that are linked to those tables (unlike Gemini-inserted charts, which are static copies), so they redraw when a dropdown changes. A third dropdown switches the monthly chart between Applications and Hires, and conditional-formatting alerts flag a low hire rate and a slow time to hire. You will hide the gridlines, protect the formulas, and ask Gemini to summarise the dashboard for the Head of Talent.

What you'll build

OrchidBay_Dashboard_Interactive: a protected Dashboard tab with three linked charts, a metric switch, KPI alerts and a Gemini summary exported to Google Docs (Tools: Google Sheets · Insert > Chart (linked) · Gemini actions (dropdown, conditional formatting) and formulas · Export to Docs · OrchidBay_Dashboard_Interactive.xlsx.)

Prerequisites

- Lab 9 completed, or its Test It values to hand.
- This lab folder downloaded, including data/OrchidBay_Dashboard_Interactive.xlsx.

Step-by-step

1. Download OrchidBay_Dashboard_Interactive.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Dashboard_Interactive (Google Sheets copy)

Dashboard : C3 Position and C4 Source dropdowns (All) | KPIs B7:H7 | tables
B10, E10, I10

Check     : B7 = 720 and E7 = 41 with All / All
```

2. Ask Gemini how to make charts that update, then build three yourself: select E10:G19 > Insert > Chart > Line chart; select B10:C14 > Insert > Chart > Bar chart; select I10:J17 > Insert > Chart > Column chart. Drag them into a grid below the tables.

```
ASK GEMINI:

How do I insert a chart from a range on this tab so that it updates when
the values in the range change?
```

3. Test interactivity. Set C3 to Customer Success Executive: the KPIs show 103 applications and 7 hires, and all three charts redraw. Set C4 to Referral, then put both back to All.

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4. Add a metric switch. Type Show in B5 and apply the dropdown action. Type ="Monthly "&C5 in H10, insert Gemini's formula in H11 and fill it down to H19. Then edit the line chart (double-click > Setup > Data range) to use E10:E19 and H10:H19. Switching C5 to Hires now redraws the line.

```
ACTION:

Create a dropdown in Dashboard!C5 with the options Applications and Hires.


FORMULA:

Create a formula for H11 that returns F11 when C5 is "Applications" and G11
when C5 is "Hires".
```

5. Add alerts, one action at a time. Test them: Operations Supervisor turns F7 red (4.3%); HR Executive turns F7 red (3.5%) and H7 amber (39.3 days).

```
ACTION:

1) Highlight Dashboard!F7 in light red when it is below 5%.

2) Highlight Dashboard!H7 in amber when it is more than 35.

3) Add a white-to-green colour scale to Dashboard!J11:J17.
```

6. Polish it yourself: View > Show > untick Gridlines; right-click the Lists tab > Hide sheet; Data > Protect sheets and ranges > protect the Dashboard tab except C3:C5, so viewers can use the dropdowns but cannot break the formulas. Then set C3 and C4 back to All and ask Gemini for a summary. Check its numbers against B7:H7, then click Export to Docs.

```
ASK GEMINI:

Summarise what the Dashboard tab shows in five bullet points for the Head
of Talent, using the KPI row and the three tables. Mention the best month,
the best source for hires and one risk.
```

7. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```


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Sample prompts

Purpose	Prompt
Image	Create an image of a simple header banner with an orchid and the words Hiring Dashboard.
Layout	Suggest a clean layout for this dashboard: where should the KPIs, charts and selectors go?
Alert	Highlight the Hires cells in the monthly table that are 0 in light grey.
Number format	Format F7 as a percentage with one decimal place and G7:H7 with one decimal place.
Review	Review this dashboard for a colleague: what is unclear, and what would you add?

Test it

Check	Expected result
Linked charts	Three charts built with Insert > Chart from E10:G19 (or E10:E19 + H10:H19), B10:C14 and I10:J17.
Customer Success Executive	103 applications, 7 hires, 6.8%; all charts redraw.
Metric switch	C5 = Hires changes the line chart and the H10 header to 'Monthly Hires'.
Alerts	Operations Supervisor: F7 red (4.3%). HR Executive: F7 red (3.5%), H7 amber (39.3).
Polish	No gridlines, Lists hidden, Dashboard protected except C3:C5; summary exported to Docs.

Checkpoint and rejoin point

Lab 11 moves to the final decision: 16 finalists for four roles, screened with a Gem.

Troubleshooting

If this happens	Fix
The chart doesn't change when you change C3.	It was inserted from the Gemini panel (a static copy) or from the wrong range. Delete it and use Insert > Chart on the Dashboard table.
The line chart shows Applications and Hires together.	Edit the data range to E10:E19 and H10:H19 only.
The F7 alert never turns red.	F7 holds a fraction (0.043), not 4.3. The rule must be 'less than 0.05' - open Settings or Format > Conditional formatting to check.
Viewers can't use the dropdowns after protection.	In Protect sheets and ranges, tick 'Except certain cells' and add C3:C5.

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Challenge

Add a 'Data as at' note in A2 that comes from a formula (e.g. ="Data as at "&TEXT(MAX(Applications!C2:C721),"yyyy-mm-dd")).

Reflection

Which three people would use this dashboard, and what would each of them need protected or hidden?

Note: The lab folder labs/lab-10-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Dashboard_Interactive.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Concepts for Lab 11: Gems: Reusable Expert Assistants, Writing Gem Instructions That Are Fair.

Gems: Reusable Expert Assistants

A Gem is a custom version of Gemini with saved instructions and, optionally, knowledge files. You create it at gemini.google.com > Gems > New Gem, test it in the preview, save it, and then use it in the Gemini app or from the Gems list in the Sheets side panel. Premade Gems are also available; Gems are not supported in every language.

Repeat tasks done by different people drift. A Gem applies the same instructions, rubric and output format every time, so screening is faster and more consistent - while the decision stays with a person.

How it works

- Write instructions with Persona, Task, Context and Format.
- Add knowledge files from Drive (job descriptions, rubric).
- Test on a known example, save, then use it from the Sheets side panel.

Worked example

- 'Orchid Bay Candidate Screener' scores finalists 0-4 on five criteria.
- Knowledge: OrchidBay_Job_Descriptions and OrchidBay_Screening_Rubric.
- Tested on one candidate: 12/20, Hold.

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Decision guide

Use when	Avoid when
The same expert task is repeated by several people.	Instructions or files contain confidential data you are not allowed to share with AI.
The rules can be written down and checked.	The Gem's recommendation would be used as the decision.

Practitioner quality lens

- Instructions: Persona, Task, Context, Format.
- Knowledge: Current, approved files only.
- Testing: Known example checked before use.

Authoritative references

- https://support.google.com/gemini/answer/15146780
- https://support.google.com/docs/answer/14356410

Writing Gem Instructions That Are Fair

Google's prompt guidance recommends four parts: Persona (who Gemini should be), Task (what to do), Context (the facts, rules and limits) and Format (how to present the answer). For HR use, Context also carries the fairness rules: which evidence to use, which characteristics never to use or guess, what to do when information is missing, and that a person makes the decision.

Instructions are the Gem's policy. Explicit evidence and fairness rules reduce bias and make the output explainable; a fixed format makes it easy to insert and check in Sheets.

How it works

- State the evidence allowed and the scoring rules.
- List the characteristics never to use or infer.
- Fix the output table and ask for an 'unsure' list.

Worked example

- Evidence: skills, experience, interview score, salary, notice period.
- Never: age, gender, race, religion, nationality, marital status, health.
- Ask the Gem which information it used for each recommendation.

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Decision guide

Use when	Avoid when
Any Gem that touches decisions about people.	Vague instructions such as 'pick the best candidate'.
Output will be inserted into a sheet and checked.	Using a Gem to process data your policy doesn't allow.

Practitioner quality lens

- Evidence: Allowed inputs listed.
- Fairness: Prohibited inputs listed.
- Transparency: Reasons and 'unsure' cases shown.

Authoritative references

- https://support.google.com/docs/answer/15013615
- https://support.google.com/gemini/answer/15146780

Lab 11 · Gems — Create a Candidate Screener Gem and Use It in Sheets

Learning outcome: LO3: create a custom Gem with instructions and knowledge files, use it from the Gemini side panel in Sheets to screen candidates consistently, and check its output for accuracy and fairness.

Goal: Build a reusable Candidate Screener Gem that scores finalists against Orchid Bay's rubric the same way every time - and check it against the data.

Screening 16 finalists by hand takes the recruiters a day, and every recruiter scores a little differently. You will create a Gem - a custom Gemini with saved instructions - in the Gemini app. Its instructions follow the Persona, Task, Context and Format pattern; its knowledge is the job descriptions and the screening rubric from Google Drive. You will test it on one candidate, open it from the Gems list in the Sheets side panel, screen all four roles, insert the results, check them against the data and record the human decision.

What you'll build

An 'Orchid Bay Candidate Screener' Gem with instructions and two knowledge files, and OrchidBay_Finalists with a Gem_Screening tab, checks and a Hiring_Manager_Decision dropdown (Tools: Gemini app (gemini.google.com) · Gems · Google Drive · Google Sheets Gemini side panel (Gems) · OrchidBay_Finalists.xlsx.)

Prerequisites

- Gems available on your account (gemini.google.com > Gems). Gems are not available in every language.

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- This lab folder downloaded: data/OrchidBay_Finalists.xlsx, the two .docx files in data/gem-knowledge/ and gem-instructions.txt.

Step-by-step

1. Upload OrchidBay_Finalists.xlsx and the two .docx files in data/gem-knowledge/ to Google Drive. Open the workbook and choose File > Save as Google Sheets. Open each .docx and choose File > Save as Google Docs, so the Gem can read them as Docs.

```
Required state:

Drive     : OrchidBay_Finalists (Sheets) | OrchidBay_Job_Descriptions
(Docs) | OrchidBay_Screening_Rubric (Docs)

Finalists : 16 rows - four each for Data Analyst, Software Engineer,
Customer Success Executive, Finance Associate
```

2. In a new browser tab go to gemini.google.com > Gems (in the left menu, or Explore Gems) > New Gem. Name it Orchid Bay Candidate Screener. Paste the instructions below (also in gem-instructions.txt) into Instructions. Under Knowledge, click Add files and choose the two Google Docs from Drive.

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```
GEMINI APP:

PERSONA

You are the Orchid Bay Candidate Screener, a careful and fair recruitment
analyst at Orchid Bay Talent Partners, a Singapore recruitment and HR firm.


TASK

Screen candidates for a role against the job description and the screening
rubric in your knowledge files, and recommend Advance, Hold or Decline for
each one.


CONTEXT

- Use only two sources: the candidate's row (Years_Experience, Key_Skills,
Expected_Salary, Budget_Max, Notice_Weeks, Final_Interview_Score,
Candidate_Summary) and the job description for the candidate's Position.

- Score each rubric criterion from 0 to 4 exactly as the rubric defines it,
then add them up to a total out of 20.

- A must-have skill counts only if it appears in Key_Skills or
Candidate_Summary.

- Never use or guess age, gender, race, religion, nationality, marital
status or health. Do not treat a name or a school as evidence of ability.

- If information is missing, say "not enough information". Do not guess.

- Your recommendation supports the hiring manager. A person makes the
hiring decision.


FORMAT

Return one table with these columns: Candidate_ID, Position, Must-have
gaps, Skills, Experience, Interview, Salary fit, Availability, Total /20,
Recommendation, Reason (one line, evidence only).

After the table, list any candidate where you were unsure and why.
```

3. Test the Gem in the preview panel on the right before saving. Paste this one candidate. A correct answer finds the missing must-have skill (Tableau) and scores 12/20 - Skills 2, Experience 3, Interview 1, Salary 4, Availability 2 - so it recommends Hold. If it is wrong, improve the instructions and test again. Then click Save.

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```
GEMINI APP:

Screen this candidate for the Data Analyst role:

CAN-26693 | Data Analyst | 4 years | Key_Skills: Excel; Google Workspace;
SQL; Stakeholder Management | Expected_Salary S$4,700 | Budget_Max S$6,500
| Notice_Weeks 8 | Final_Interview_Score 3.2
```

4. Go back to OrchidBay_Finalists in Sheets. Open the Gemini side panel, click Gems and choose Orchid Bay Candidate Screener (reload the sheet if it isn't listed yet). Screen the first role and click Insert, putting the table on a new tab named Gem_Screening.

```
GEM:

Screen the four Data Analyst finalists on the Finalists tab using the
rubric. Identify candidates by Candidate_ID only and ignore Full_Name.
Return the table in your standard format.
```

5. Screen the other three roles. Insert adds each table on a new tab - keep them, or use Copy and paste each one onto Gem_Screening below the Data Analyst table, leaving a blank row between tables.

```
GEM:

Now screen the Software Engineer, Customer Success Executive and Finance
Associate finalists in the same format.
```

6. Check the Gem against the data. Must-have gaps should be exactly: CAN-26693 (Tableau); CAN-26699 (JavaScript); CAN-26622 (JavaScript); CAN-26621 (Reconciliation); CAN-26579 (Accounting). Above budget: CAN-26567, CAN-26577, CAN-26579, CAN-26662. Applying the rubric exactly gives 7 Advance, 8 Hold and 1 Decline. Paste the check and list any row where the Gem differs - then ask the Gem about its evidence.

```
CHECK:

Above budget   =SUMPRODUCT(--(Finalists!F2:F17>Finalists!G2:G17))


GEM:

For each recommendation, which pieces of information did you use? Did you
use anything other than skills, experience, interview score, salary and
notice period?
```

7. Record the human decision - the Gem only recommends. Apply the action, then choose a decision for the Data Analyst rows yourself.

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```
ACTION:

Add a column called Hiring_Manager_Decision at the end of the Data Analyst
table on the Gem_Screening tab, with a dropdown of Advance, Hold and
Decline.
```

8. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```

Sample prompts

Purpose	Prompt
Gem	Write three interview questions for the weakest rubric area of each Advance candidate.
Gem	Rank the Software Engineer finalists by Total and explain the ranking in one sentence each.
Gem	Draft a polite email to a candidate you recommend we Decline, without giving scores.
Gemini app	Improve my Gem instructions so the Reason column always quotes the evidence.
Side panel	What are the main points of OrchidBay_Screening_Rubric?

Test it

Check	Expected result
Gem	Saved as 'Orchid Bay Candidate Screener' with instructions and both knowledge files.
Test candidate	CAN-26693: gap Tableau; 12/20; Hold.
Must-have gaps	CAN-26693 (Tableau); CAN-26699 (JavaScript); CAN-26622 (JavaScript); CAN-26621 (Reconciliation); CAN-26579 (Accounting).
Salary check	Above-budget formula = 4 (CAN-26567, CAN-26577, CAN-26579, CAN-26662).
Decisions	Rubric applied exactly: 7 Advance / 8 Hold / 1 Decline; Hiring_Manager_Decision dropdown added and filled by you.

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Checkpoint and rejoin point

Lab 12 uses the same 16 finalists to choose the hires within the client's budget.

Troubleshooting

If this happens	Fix
The Gem isn't listed in the Sheets side panel.	Reload the sheet. If Gems are not available in the side panel on your account, use the Gem at gemini.google.com and paste only the screening columns - Candidate_ID and C1:L17, never Full_Name - into the chat.
The Gem can't find the rubric.	Check that both files were added under Knowledge and are Google Docs you can open.
The scores don't follow the rubric.	Add to the instructions: 'Show the evidence for each criterion score before the total.' Test again.
The Gem mentions a candidate's background or school.	That breaks the fair-screening rules. Strengthen the Context section and report it with the thumbs-down.

Challenge

Create a second Gem, 'Orchid Bay Interview Planner', that turns a Gem_Screening row into a 30-minute structured interview plan.

Reflection

What makes a Gem more consistent than typing the same long prompt each time - and what does it not fix?

Note: The lab folder labs/lab-11-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Finalists.xlsx, data/gem-knowledge/OrchidBay_Job_Descriptions.docx, data/gem-knowledge/OrchidBay_Screening_Rubric.docx), plus gem-instructions.txt (the Gem instructions to paste). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Concepts for Lab 12: The Optimization Problem Solver, Sharing and Responsible Use.

The Optimization Problem Solver

Gemini in Sheets can solve optimisation tasks - choosing the best combination under limits, such as allocating budgets, scheduling staff or selecting projects. You describe the objective (maximise or minimise something), the decisions (what can change) and the constraints (the limits that must hold).

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Hiring under a budget is a combination problem: the 'top two per role' may not fit. A solver finds the best feasible choice and shows the trade-off - and formulas prove the constraints hold.

How it works

- State the objective, the decisions and every constraint.
- Compare with the obvious choice to see the trade-off.
- Record the choice with checkboxes and prove it with SUMIFS and COUNTIFS.

Worked example

- Maximise total Weighted_Score; 2 hires per role; salary within S$41,900.
- Top two per role costs S$43,100 - over budget.
- The optimum swaps one Software Engineer and loses 1 point.

Decision guide

Use when	Avoid when
Many combinations and hard limits.	The objective leaves out what matters (e.g. team fit).
You can verify the result with formulas.	You accept the answer without checking the constraints.

Practitioner quality lens

- Model: Objective, decisions, constraints written out.
- Proof: Constraints checked with formulas.
- Judgement: A person reviews what the score misses.

Authoritative references

- https://support.google.com/docs/answer/14356410

Sharing and Responsible Use

Gemini can use your Drive files ('What are the main points of ...') and Gmail as context, and lists the Sources it used. When a file is shared outside your domain, the side panel warns that content added by Gemini may be visible to external users. Conversations in Sheets are not saved to Gemini Activity, and Clear history removes output you have not inserted. Gemini can be inaccurate and is not professional advice.

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HR data is personal data. Responsible use means minimising what goes into prompts and shared files, checking sources, keeping a human decision, and giving feedback when output is wrong or unsafe.

How it works

- Use only approved, minimal data; check the Sources of any answer.
- Share a minimal view, view-only, and heed the external-sharing warning.
- Keep the decision with a person; give feedback; clear history.

Worked example

- Share_View holds only Candidate_ID, Position, Weighted_Score and Selected.
- The recommendation is exported to Docs and checked against the formulas.
- Fictional data in class; real data only where your policy allows.

Decision guide

Use when	Avoid when
Before sharing any sheet that holds personal data.	Pasting confidential or personal data into a tool your organisation hasn't approved.
Whenever Gemini uses Drive, Gmail or the web.	Letting AI output stand as the decision about a person.

Practitioner quality lens

- Minimise: Only the fields the viewer needs.
- Source: Files and web sources checked.
- Accountable: A person decides and signs off.

Authoritative references

- https://support.google.com/docs/answer/14356410
- https://support.google.com/docs/answer/1218656

Lab 12 · Optimisation, Reporting & Safe Sharing — Optimise the Hiring Decision and Share Results Safely

Learning outcome: LO3: use Gemini's optimisation problem solver to choose hires under constraints, verify the solution with formulas, export a recommendation to Google Docs, use Drive files as context and share HR results responsibly.

Goal: Choose the eight hires that give the highest total Weighted_Score within the client's salary budget, prove the answer, and share only what the client needs.

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The client wants eight hires - two for each of four roles - and one total monthly salary budget of S$41,900. Picking the top two in each role costs S$43,100, which is over budget. You will describe the problem to Gemini's optimisation solver, compare its answer with the no-budget choice, tick the selected candidates and prove the totals with formulas. Then you will export a recommendation to Google Docs, use a Drive file as context, build a minimal Share_View tab and share it safely.

What you'll build

OrchidBay_Hiring_Decision with a Selected checkbox column, check formulas that prove the constraints, a recommendation in Google Docs and a separate Share_View file shared view-only (Tools: Google Sheets · Gemini optimisation problem solver · Export to Docs · Drive files as context · sharing · OrchidBay_Hiring_Decision.xlsx.)

Prerequisites

- Lab 11 completed (the rubric Doc is in your Drive), or upload OrchidBay_Screening_Rubric.docx from the Lab 11 folder.
- This lab folder downloaded, including data/OrchidBay_Hiring_Decision.xlsx.

Step-by-step

1. Download OrchidBay_Hiring_Decision.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

```
Required state:

File      : OrchidBay_Hiring_Decision (Google Sheets copy)

Tabs      : About_This_Data | Finalists (16 rows) | Constraints |
Prompt_Log

Budget    : Constraints!B8 = S$41,900 per month for 8 hires
```

2. Describe the optimisation problem in plain English: the goal, what can change, and the limits. Gemini's optimisation solver works out the best combination. Note the candidates it picks, their total Weighted_Score and total salary - you prove them in Step 4 and compare them with Test It.

```
ASK GEMINI:

I need to choose hires from the Finalists tab. Maximise the total
Weighted_Score. Hire exactly the number of Openings for each Position on
the Constraints tab, and keep the total Expected_Salary of everyone hired
at or below the budget in Constraints!B8. Which candidates should I hire?
```


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3. Compare with the obvious choice. Without the budget, the top two per role cost S$43,100 - over the S$41,900 budget - so something has to give. Ask Gemini what it traded and how many points that cost. Note: the optimiser uses Weighted_Score only; remove anyone your Lab 11 review marked Decline before you rely on it.

```
ASK GEMINI:

If there were no budget limit, which candidates would you pick, what would
they cost, and what is the trade-off compared with your answer?
```

4. Add the checkbox column with an action, then tick the eight recommended candidates yourself. Paste the checks into Constraints D2:D7 - they must show the budget is met and two hires per role.

```
ACTION:

Add a checkbox column called Selected at the end of the table on the
Finalists tab, unchecked by default.
```

```
CHECK:

Total salary     =SUMIFS(Finalists!F2:F17,Finalists!M2:M17,TRUE)

Total score      =SUMIFS(Finalists!K2:K17,Finalists!M2:M17,TRUE)

Within budget    =SUMIFS(Finalists!F2:F17,Finalists!
M2:M17,TRUE)<=Constraints!B8

Data Analysts    =COUNTIFS(Finalists!C2:C17,"Data Analyst",Finalists!
M2:M17,TRUE)

All hires        =COUNTIF(Finalists!M2:M17,TRUE)
```

5. Draft the recommendation, check every number against your formulas, then click Export to Docs. Next, use a Drive file as context - expand Sources under the answer to confirm which file Gemini used.

```
ASK GEMINI:

Write a hiring recommendation for the client in under 200 words: the eight
recommended hires by role, the total salary against the budget, the one
trade-off made because of the budget, and the risks to check (notice
periods and any must-have gaps).
```

```
ASK GEMINI:

What are the main points of OrchidBay_Screening_Rubric? Does my
recommendation follow its fair screening rules?
```


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6. Share only what the client needs. Ask Gemini which columns are personal data the client doesn't need, then create a minimal view. Do not share this workbook - it still holds names and summaries. Instead right-click the Share_View tab > Copy to > New spreadsheet, open the new file, and share only that file with your trainer as Viewer. Watch for the warning that content may be visible outside your domain.

```
ASK GEMINI:

Before I share this spreadsheet with the client, which columns contain
personal data they don't need to see?


ACTION:

Create a new tab called Share_View with only Candidate_ID, Position,
Weighted_Score and Selected for the rows where Selected is checked.
```

7. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

```
LOG:

List the prompts I sent you in this conversation. For each one, give a one-
line summary of what you did, so I can copy it into the Prompt_Log tab.
```

8. Close the loop: give feedback on one helpful and one unhelpful answer with the thumbs up or down. Then, after Prompt_Log is complete, open More options > Clear history to remove the side-panel conversation.

Sample prompts

Purpose	Prompt
Optimise	What is the highest total Weighted_Score possible if the budget is reduced by S$2,000?
Optimise	Schedule the eight final interviews across Monday and Tuesday so no interviewer has more than three in a row.
Docs	Turn the recommendation into a one-page memo with a table of hires and start dates.
Gmail	Catch me up on the latest emails about the Q3 hiring drive.
Share	Which tabs in this spreadsheet would be risky to share outside Orchid Bay, and why?

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Test it

Check	Expected result
Optimum	CAN-26567, CAN-26608, CAN-26621, CAN-26627, CAN-26683, CAN-26692, CAN-26699, CAN-26711.
Totals	Total score 673.5; total salary S$40,700 (budget S$41,900); Within budget = TRUE.
Per role	Two Selected per role; All hires = 8.
No-budget comparison	674.5 for S$43,100; swap CAN-26622 -> CAN-26627.
Share safely	Recommendation exported to Docs; Share_View has 4 columns and 8 candidate rows (plus a header); only the copied Share_View file is shared, as Viewer.

Checkpoint and rejoin point

This is the final lab. Your Drive now holds the full Orchid Bay workflow from raw export to decision.

Troubleshooting

If this happens	Fix
Gemini answers without solving - it just picks the top scores.	Restate the constraints as rules ('exactly 2 per Position', 'total salary at or below S$...'), or ask: 'Use the optimisation solver.' The solver may not be available on every plan - then check the answer with the CHECK formulas.
Gemini's answer breaks the budget.	Tick its picks and look at Within budget - it will show FALSE. Tell Gemini the total and ask again.
The SUMIFS show 0.	The Selected column is not M, or the boxes are not ticked. Check the column letter in the formula.
No optimisation solver on your plan.	Find it by hand: ask Gemini to 'list every pair of candidates for each Position with their combined Weighted_Score and Expected_Salary'. Start from the top pair in each role and swap the pair that loses the fewest points per S$ saved until the total salary fits the budget.
Gemini can't find the rubric file.	Upload OrchidBay_Screening_Rubric.docx from the Lab 11 folder to Drive and save it as a Google Doc.

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Challenge

Ask Gemini for the best selection if the client adds a third Data Analyst opening and S$5,000 to the budget. Prove it with the checks.

Reflection

The optimiser maximised a score. What does the score NOT capture that a hiring manager must still judge?

Note: The lab folder labs/lab-12-*/ holds the instruction PDF with sample prompts, sample-prompts.txt and the mock-data workbook (data/OrchidBay_Hiring_Decision.xlsx). Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Topic 03 Recap - Mapped to Learning Outcomes

Use this checkpoint to explain what you can now do and identify any lab evidence you still need to repair.

Learning outcome	Evidence from this topic
LO3 · KPI formulas	Dropdown selectors drive COUNTIFS and AVERAGEIFS KPIs and summary tables.
LO3 · Interactivity	Linked charts, a metric switch, KPI alerts and protection - tested on three views.
LO3 · Gems	A Candidate Screener Gem with instructions and knowledge files, used from the Sheets side panel.
LO3 · Decide and share	Optimised hires within budget, proved with formulas, exported and shared minimally.

Wrap-Up - The Complete C817 Workflow

Across twelve labs you took one hiring drive from a raw export to a decision: cleaned, screened, tracked, analysed, visualised, put on an interactive dashboard, screened with a Gem and optimised within budget - with a formula check between every Gemini output and every people decision.

Fundamentals of Google Sheet

- Open Gemini on a native Sheets file; name tabs and columns; insert what you want to keep.
- Profile and flag data-quality problems, then prepare the data with actions you preview and undo.

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- Describe formulas in plain English; read the explanation; fix errors with Fix.

Data Analytics and Visualization

- Turn notes into trackers with dropdowns, checkboxes and conditional formatting.
- Use =AI and Fill with Gemini for text - and measure the accuracy.
- Ask for insights and pivot tables, read Analysis steps, and chart with titles that state the finding.

Interactive Dashboard

- Drive every KPI and table from dropdowns with COUNTIFS and AVERAGEIFS.
- Use linked charts, alerts and protection; summarise and export to Docs.
- Automate repeat screening with a fair Gem; optimise within constraints; share the minimum.

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Next Steps

- Re-run all twelve labs from the original workbooks without looking at your finished files.
- Pick one weekly spreadsheet task at work and write the Gemini prompt for it - then check the result with a formula.
- Build one interactive dashboard for a report you produce every month.
- Create one Gem for a repeat task, with fair, evidence-only instructions.
- Check Google's Collaborate with Gemini in Google Sheets page regularly - features and labels change often.

Glossary

- **Action —** A change Gemini makes to the sheet from a prompt, shown on an Action preview card and applied with Apply.
- **Action preview card —** The card that describes a proposed Gemini action before you apply it.
- **AI function —** =AI("prompt", [range]) - a Sheets function that sends a prompt to Gemini and returns text into a cell.
- **Analysis steps —** A control under a Gemini answer that shows how it analysed the data.
- **AVERAGEIFS —** Averages a range for rows that meet one or more conditions.
- **Conditional formatting —** Formatting that applies when a rule is true, e.g. highlight rows where score >= 85.
- **COUNTIF / COUNTIFS —** Count cells that meet one condition / rows that meet several conditions.
- **Dropdown —** An in-cell list of allowed values.
- **Export to Docs —** Sends a Gemini response to a new Google Doc.
- **Fill with Gemini —** Completes a column from examples or a prompt, using the context of the table.
- **Fix —** Gemini's button on a formula error that explains and corrects the formula.
- **Gem —** A custom Gemini with saved instructions and knowledge files, reusable in the Gemini app and side panel.
- **Hire rate —** Hires divided by applications for the same group.
- **Insert —** Puts a Gemini table, text, formula or chart into the sheet.
- **KPI —** Key performance indicator: one headline measure calculated by formula.
- **Linked chart —** A chart built with Insert > Chart on a live range, so it updates when the range changes.
- **Optimization problem solver —** Gemini's solver for choosing the best combination under constraints.
- **Pivot table —** A summary that groups rows and aggregates values by fields in rows, columns and filters.
- **Prompt —** The instruction you give Gemini; best with task, tab and column names, output format and rules.

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- **Sources —** The tabs, files or web pages Gemini used; choose tabs in the prompt box and expand Sources under answers.
- **VLOOKUP —** Finds a key in the first column of a range and returns a value from another column.
- **Wildcard * —** In COUNTIFS criteria, matches any text - used so 'All' counts every value.

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